

Planar matching under monotone pairwise angles

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Results obtained: 6–7 and 23 September 2026

Manuscript: 23 September 2026 · Version 3.1.0

Abstract

We bound the bijections that become uniquely optimal for squared-distance matching along a planar motion. Suppose the labeled targets are periodic and collision-free, with every pairwise relative angle nonincreasing. If $\mathcal{W}$ is their total clockwise pair winding, at most $\max(1, 2\mathcal{W})$ bijections occur. The same bound holds for a coordinate-lexicographic canonical selection, even with repeated sources. Assignment duality, monotone interpolation of finite velocity data, and a planar flux identity yield positive transition charges with generic sum $2\mathcal{W}$; no common velocity field is required. Rigid rotation has the sharp count $n(n-1)$, also attained by nonrigid motions with unit pair windings. For densely encoded rational Fourier trajectories, we check the geometric hypotheses and solve registration exactly in polynomial bit time, without a genericity promise. The output covers the optimal value rather than all tied assignments. Counterexamples show factorial diversity at fixed winding without angular monotonicity, and thirteen matchings for four points under fixed-axis three-dimensional rotation. Further results give polynomial counts for specified small mass perturbations in self-registration, with a sharp linear bound for one defect between hull terminals on positive-cycle intervals.

Contribution and significance

Zikan gave an exact parametric registration method; Rote subsequently asked whether the number of optimal-matching intervals under rotation admits a polynomial bound. We prove the sharp rotational count $n(n-1)$ and a trajectory-level theorem governed by pair winding. The proof links an assignment dual to a convex piecewise-affine potential, extends finite monotone velocity data by a classical interpolation theorem, and uses planar flux to orient each transition. Positive local charges then turn topological motion into a quantitative matching bound.

An interval certificate detects hidden switches even when endpoint assignments agree. Together with source-coordinate lexicographic selection and algebraic root separation, it gives a checked exact Fourier algorithm for degenerate inputs. Counterexamples separate angular monotonicity from final winding data and exhibit a dimensional obstruction. The transport appendices supply restricted positive extensions through Delaunay support and monotone face potentials. The contribution is the structured bounds and their certificates, not the classical assignment, interpolation, braid, or algebraic tools. Section 2 gives the source comparison and its limitations.

1. Introduction

For a fixed motion, least-squares matching is an assignment problem; for a fixed bijection, rigid registration has an explicit solution. Neither observation controls how many bijections can be optimal during a motion. The geometric structure of the cost matrix must be used: a low-dimensional parameter alone does not give the bounds proved here.

We identify the oriented plane with $\mathbb{C}$ and also use real column vectors. Let $J(x, y) = (-y, x)$, and let R_θ denote counterclockwise rotation. Given fixed sources $A = (a_1, \dots, a_n)$ and moving targets $B(t) = (b_1(t), \dots, b_n(t))$, minimizing

$$\sum_i |a_i - b_{\pi(i)}(t)|^2$$

is equivalent to maximizing $S_\pi(t) = \sum_i a_i \cdot b_{\pi(i)}(t)$. The target norm sum is independent of π , even for nonrigid motion. A permutation is counted only if it is the unique optimum at some time. Counting every permutation participating in a tie would allow $n!$ even for constant point lists.

Our principal theorem, [Theorem 3.1](#), bounds this count by the total pair winding under monotone pairwise angles. Its proof is complete in [Section 3](#); the generic event identity is stronger than the all-source distinct-permutation count. Finite-witness perturbation also controls a canonical selection and its open cells. Rigid rotation gives the sharp bound in [Theorem 3.6](#). Polynomial shear conjugacies show that the mechanism is not confined to rigid motion; a breathing example shows why convex Hamiltonian interpolation is not necessary.

[Section 4](#) makes the topological budget algorithmic. The squared Vandermonde along a straight chord between canonically reordered endpoint configurations, followed by the reversed physical motion, gives a computable nonnegative integer. Its vanishing certifies absence of canonical switches. Uniform separation of algebraic candidate times gives polynomial-depth subdivision without generating the factorial family of score differences. For Fourier degree D , total charge is at most $n(n-1)D$; at unit pair winding the count remains $n(n-1)$ while runtime still depends on D .

[Section 5](#) tests the necessity and dimensional boundary of the hypotheses. The appendices retain sharper rigid and common-metric algorithms, angular-envelope consequences, and unequal-mass transport extensions. Throughout, $n = 1$ is handled separately.

The weighted contrast is not solely negative. In self-registration near the identity phase, a small binary perturbation of balanced masses reduces to a minimum-cost flow with edge energy equal to squared length minus an oriented-area term. Every uniquely optimal flow uses strict Delaunay edges. Integer face potentials turn the area parameter into a uniform monotone forcing of a submodular energy. This gives the cubic count in [Theorem D.6](#), the quadratic one-defect bound in [Theorem D.5](#), and the linear hull-terminal bound in [Theorem D.8](#). A finite-amplitude counting argument admits arbitrary real mass amplitudes in [Theorem D.7](#). An explicit parabolic fan attains the hull-terminal linear bound for every size on a certified positive-cycle interval, as shown in [Theorem D.11](#). These are restricted positive results, not a solution of arbitrary-mass rotational transportation. They do not enter the proof of the winding theorem or the Fourier algorithm.

2. Relation to earlier work and contribution boundaries

Exact planar least-squares registration with unknown correspondence predates the present work. Zikan's report [1] and abbreviated journal article [2] develop the complex-coordinate assignment formulation. The journal article proves a convex conical subdivision theorem and enumerates representatives of its two-dimensional cells, optimizing the transformation for each representative. Its discussion on p. 170 explicitly leaves worst-case complexity unresolved; the approximately quadratic pivot count is empirical. Our contribution is therefore neither first exact solvability nor the conical subdivision. The sharp rotational count supplies a worst-case polynomial guarantee through a separate vertex-returning assignment oracle and output-sensitive enumeration; it does not bound degenerate or blocked pivots of Zikan's simplex implementation.

Rote's Oberwolfach problem [3] asks whether the number of rotational optimality intervals is polynomially bounded, alongside questions about local minima and globally optimal rotation.

Problem 4 of the subsequent Dagstuhl session [4] records the counting question again and contrasts the geometric setting with unrestricted parametric assignment. Neither statement prescribes the exact value $n(n-1)$. We distinguish permutations that are uniquely optimal somewhere from all permutations participating in ties; the latter can be factorially numerous. The sharp value and its all-size realization are the refinement established here, not the wording of the historical question. In particular, Rote’s Note (1) on p. 2547 describes the quadratic behavior as empirical, not as a proved bound.

Subsequent approximation work must also be distinguished from exact enumeration. Utriainen’s thesis [5] studies least-squares rigid fitting with unknown bijections in two and three dimensions. Its abstract describes approximation schemes for a vector-weighted bipartite matching formulation and warns that the relative guarantee need not transfer to the original squared-error objective. That stated guarantee is different from an exact worst-case count of rotational optimality cells. We cite this abstract-level comparison without treating it as a complete reading of the thesis.

The algorithmic ingredients must likewise be separated from the geometric bound. The Hungarian method is classical. Bicriteria scalarization and dichotomic enumeration go back to work including Aneja and Nair [6]; Bökler et al. [7] give an output-sensitive analysis used here for the planar correlation polygon. These methods are not claimed as new generic hull algorithms. Ben-Avraham et al. [8] bound partial-matching translation regions crossed by a convex path by $m(n-m)+n(n-1)$. Their translation parameter and selected target subset differ from full-bijection rotation. The common numerical term does not identify the two theorems.

Weighted transportation introduces another important boundary. Cabello et al. [9] study Euclidean Earth Mover’s Distance, with unsquared ground cost and arbitrary masses, and obtain approximation algorithms under rotations and rigid motions. Their sampled angular sets are not a count of exact optimal transportation bases. Ding, Liu and Ye [10] do use squared Euclidean transportation costs, but their compression theorem transfers approximate rigid alignments with multiplicative and diameter-dependent additive error. Neither inspected result supplies the exact event count considered here. Our unequal-mass example also shows why the original equal-weight bound cannot simply be carried over to arbitrary transportation marginals.

The nonrigid extension changes the motion while retaining bijective matching. Ni et al. [18] study capacitated kinetic clustering with prescribed integer capacities and moving terminals and stations. Their certificate count is $n \binom{n}{k-1} g(k-1)$ in their notation, under bounded pseudo-algebraic event multiplicity. Its fixed- k interpretation does not give a polynomial bound when both supports have size n , or a degree-independent count at unit pair winding. We instead restrict pair angular motion and charge actual assignment changes.

The topological invariant itself is classical. In braid Floer theory, van den Berg et al. [19] express algebraic crossing number as twice the sum of pair windings; its monotone evolution there concerns Floer trajectories, not matching along physical time. Morabito [20] records, following Le Calvez, linking jumps along generating-function gradient lines. Neither result is an assignment count. At unit winding our motion represents a full twist by the positive-braid characterization attributed to Casson by Elrifai and Morton [21, Section 5]; the reduction in [Corollary 3.5](#) is an application, not a new braid classification.

The finite-data bridge also uses established extension theory. Kirszbraun’s theorem preserves a Lipschitz constant under Hilbert-space extension; see [16]. Approximate interpolation of finite monotone data appears explicitly as Lemma 1 in the cable-web analysis of Bouchitté et al. [17]. We include the short Cayley-transform argument needed here. The assignment-specific step is that the dual polygon’s flux depends only on vertex velocities, so the extension can be used at one time and then removed by a limit. This yields a positive explicit local charge, its sharp global budget, and an interval certificate. Assignment duality, Green’s formula, transversality [11], and the extension

theorem are not claimed as new.

Turning that count into an algorithm requires an additional argument. Guibas’s kinetic-data-structure framework [12] distinguishes external changes of the maintained object from internal certificate failures. A static assignment algorithm’s comparisons need not yield an efficient kinetic structure. Gassner and Klinz [13] give a fast exact parametric assignment algorithm for costs $c_{ij} - b_{ij}\lambda$ with $b_{ij} \in \{0, 1\}$, not general polynomial objective curves. Geometric exploration of multiparametric linear programs [14] is another relevant precedent, but its inspected description does not give the guarantee needed here. Our interval certificate uses the positive geometric charge to exclude actual matching switches; the accompanying separation argument controls subdivision depth. Exact real-root isolation [15], algebraic sign decisions and algebraic minimization remain classical ingredients.

The comparison is bounded by the cited problem statements and theorem passages. In particular, the technical-report predecessor and the relevant journal passages on pp. 169–171 have been examined, but the latter were available as indexed text rather than a complete visual reading of the four-page article. Some other references were consulted only through abstracts or selected sections, as noted above. A targeted search for subsequent exact rotational-count and monotone-angular-motion results did not identify a duplicate of the stated theorems; this is not a comprehensive forward-citation or priority survey. The contribution claims are the specific comparisons above, not an assertion that every later treatment has been excluded. Neither these comparisons nor the supplied computations constitute independent mathematical validation.

3. Winding and matching complexity

Let $b_1, \dots, b_n : \mathbb{R}/T\mathbb{Z} \rightarrow \mathbb{R}^2$ be smooth labeled trajectories, with $b_i(t) \neq b_j(t)$ whenever $i \neq j$. Assume

$$\det(b_j(t) - b_i(t), b'_j(t) - b'_i(t)) \leq 0 \quad (i < j, t \in \mathbb{R}/T\mathbb{Z}). \quad (3.1)$$

Thus every pair direction turns monotonically clockwise, possibly with stationary intervals. No common velocity field or potential is assumed. For a nonvanishing complex path, $\Delta \arg$ always denotes a continuous lift of argument. Define

$$w_{ij} = -\frac{1}{2\pi} \Delta \arg(b_j - b_i), \quad \mathcal{W} = \sum_{i < j} w_{ij}, \quad N = \max(1, 2\mathcal{W}). \quad (3.2)$$

Theorem 3.1 (Winding bound). *For $n \geq 2$ and any fixed source list A , at most N different permutations are uniquely optimal at some time in one period under (3.1). Each w_{ij} is a nonnegative integer. For a generic source list, every optimal tie is isolated and transverse, with exactly two optimal permutations differing by one nontrivial cycle. If k_e is its length and, when $k_e \geq 3$, r_e counts the targets outside that cycle lying inside its directed transition polygon at the switching time, then*

$$\sum_e q_e = 2\mathcal{W}, \quad q_e = \begin{cases} 1, & k_e = 2, \\ k_e - 1 + 2r_e, & k_e \geq 3. \end{cases} \quad (3.3)$$

In particular the number E of switches satisfies

$$E = 2\mathcal{W} - \sum_{e: k_e \geq 3} (k_e - 2 + 2r_e) \leq 2\mathcal{W}.$$

For arbitrary sources, the full source-coordinate lexicographic policy defined below has at most N distinct values, including values selected only at a tie point. It is constant on at most N open cells whose closures cover the original optimal value. This does not enumerate all tied permutations or permit arbitrary tie-breaking.

The trivial ceiling $n!$ can be imposed on the distinct count. For $n = 1$ the sole policy is constant, consistent with $N = 1$ rather than the empty pair-winding total. The differential condition (3.1), not merely the final winding numbers, is required.

3.1. Angular monotonicity and generic sources

For $d = b_j - b_i$, the hypothesis gives

$$\frac{d}{dt} \arg d = \frac{\det(d, d')}{|d|^2} \leq 0. \quad (3.4)$$

Periodicity makes the total argument increment an integral multiple of 2π , so $w_{ij} \in \mathbb{Z}_{\geq 0}$.

Here and below, generic sources mean an open dense subset of $\mathbb{R}^{2n}$ determined by the fixed target motion. For a permutation π , write $C_\pi(t) = (b_{\pi(1)}(t), \dots, b_{\pi(n)}(t)) \in \mathbb{R}^{2n}$. These vectors are distinct and have the same norm. Hence no three distinct ones are collinear, since a line meets a sphere in at most two points. For $\pi \neq \tau$, the function $f(A, t) = A \cdot (C_\tau(t) - C_\pi(t))$ is a submersion in A . For three distinct permutations, the two corresponding differences are linearly independent, so the two-component tie map is also a submersion. The classical parameter transversality theorem [11, Theorem 1.1] implies that, for generic A , all pair ties are transverse and no triple tie occurs on the one-dimensional time circle. There are finitely many pairs and triples. Transverse zeros are isolated and therefore finite on that compact circle; the simultaneous conditions form a dense open set. Only the source list is varied: neither the trajectories nor their period or pair windings are perturbed.

3.2. Finite monotone interpolation

The following classical extension mechanism, related to finite monotone interpolation in [17, Lemma 1], is included to specify exactly what is needed from the velocity data. Its only extension input is Kirszbraun's Euclidean Lipschitz extension theorem [16, Theorem 1.2].

Lemma 3.2 (Smooth approximation of finite monotone data). *Let $x_1, \dots, x_n \in \mathbb{R}^2$ be distinct and let $F_1, \dots, F_n \in \mathbb{R}^2$ satisfy $(x_i - x_j) \cdot (F_i - F_j) \geq 0$. There are smooth globally monotone maps $F^{(\ell)} : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ with $F^{(\ell)}(x_i) \rightarrow F_i$ for every i . In particular $\text{sym } DF^{(\ell)} \geq 0$ everywhere.*

Proof. Fix $\epsilon > 0$ and set $G_i = F_i + \epsilon x_i$, $u_i = x_i + G_i$, and $z_i = x_i - G_i$. Then

$$|u_i - u_j|^2 - |z_i - z_j|^2 = 4(x_i - x_j) \cdot (G_i - G_j) > 0 \quad (i \neq j).$$

The u_i are distinct, and the finite map $u_i \mapsto z_i$ is κ -Lipschitz for some $0 \leq \kappa < 1$. Extend it, with the same constant, to $Z : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ by Kirszbraun's theorem; when $\kappa = 0$ the constant extension suffices. For each x the contraction $u \mapsto 2x - Z(u)$ has a unique fixed point. Consequently

$$x = \frac{1}{2}(u + Z(u)), \quad F_\epsilon(x) = \frac{1}{2}(u - Z(u))$$

defines a global map interpolating G_i . For two arguments, denote differences by δ . We have

$$\begin{aligned} \delta x \cdot \delta F_\epsilon &= \frac{1}{4}(|\delta u|^2 - |\delta Z|^2) \geq \frac{1 - \kappa}{1 + \kappa} |\delta x|^2, \\ |\delta F_\epsilon| &\leq \frac{1 + \kappa}{1 - \kappa} |\delta x|. \end{aligned}$$

Thus F_ϵ is Lipschitz and monotone. Convolution with a nonnegative smooth compactly supported mollifier preserves monotonicity: integrate its defining inequality at $x - z, y - z$. The convolutions converge locally uniformly to F_ϵ . Choose the smoothing radius so the error at every x_i is below ϵ , and then let $\epsilon \downarrow 0$. Differentiating the monotonicity inequality along each direction yields the asserted positive-semidefinite symmetric Jacobian. No uniform Lipschitz bound as $\epsilon \downarrow 0$ is needed. $\square$

3.3. Exact dual support and the transition polygon

Fix a generic switching time t_0 and relabel the old optimum as identity and the new one as σ . Put $y_j = b_j(t_0)$. For the instantaneous score matrix $\omega_{ij} = a_i \cdot y_j$ put $d_{ij} = \omega_{ij} - \omega_{ii}$. Every directed simple cycle C has $D(C) = \sum_{(i,j) \in C} d_{ij} \leq 0$, by optimality of identity. The nontrivial cycles of σ sum to zero; each is therefore zero. There can be only one such cycle, since switching just one of two would produce a third optimum. Every other simple nontrivial zero cycle would likewise give another optimum.

Let $\mathcal{E}$ be the union of diagonal and σ edges. Every simple cycle using an edge outside $\mathcal{E}$ has strictly negative weight; otherwise it would give a third optimum. There are only finitely many simple cycles. Choose $\eta > 0$ smaller than every positive ratio $-D(C)/|C \setminus \mathcal{E}|$ for cycles using an edge outside $\mathcal{E}$. If there are none, take any $\eta > 0$. The adjusted weights $h_{ij} = d_{ij} + \eta \mathbf{1}_{(i,j) \notin \mathcal{E}}$ have no positive cycle. Add a supersource with zero-weight edges to all vertices; the maximum walk weights c_j from it are finite, since deleting nonpositive cycles does not decrease weight. They satisfy $c_j - c_i \geq h_{ij}$. With $r_i = \omega_{ii} - c_i$ we obtain

$$r_i + c_j - \omega_{ij} \geq \eta \mathbf{1}_{(i,j) \notin \mathcal{E}}.$$

Diagonal slacks vanish. The nonnegative slacks along the σ -cycle sum to zero and thus vanish separately. Consequently

$$r_i + c_j = \omega_{ij} \iff (i, j) \in \mathcal{E}. \quad (3.5)$$

The potentials are dual-optimal because their total equals the identity score.

Let $g(x) = \max_i (a_i \cdot x - r_i)$. The dual inequalities and their diagonal equalities give $g(y_j) = c_j$. At a cycle target y_j , exactly rows j and $\sigma^{-1}(j)$ are active; at a fixed target, only its own row is active. If $k \geq 3$, the two endpoint active sets of $[y_i, y_{\sigma(i)}]$ intersect only in row i . Every other affine function is strictly smaller at at least one endpoint, so the entire open edge lies in the unique open cell with gradient a_i . Different open edges cannot intersect, and no cycle vertex can lie on an open edge. Thus the edges form a simple polygon. Fixed targets lie off its boundary. If $k = 2$, the exchange segment has exactly the two cycle rows active, and no fixed target lies on it; no polygon assertion is used.

3.4. Convex flux fixes the orientation

The old-to-new score difference

$$D(t) = \sum_i a_i \cdot (b_{\sigma(i)}(t) - b_i(t))$$

has $D(t_0) = 0$ and $D'(t_0) > 0$. First consider any smooth test field $v = (v_1, v_2)$ satisfying $S = \text{sym}(JDv) \geq 0$. For a long cycle polygon P , edgewise integration gives

$$\sum_i a_i \cdot (v(y_{\sigma(i)}) - v(y_i)) = \oint_P (\nabla g)^T Dv \, dx. \quad (3.6)$$

For smooth g , direct differentiation of this one-form gives

$$\begin{aligned} \text{curl}((\nabla g)^T Dv \, dx) &= (v_1)_y g_{xx} + ((v_2)_y - (v_1)_x) g_{xy} - (v_2)_x g_{yy} \\ &= \text{tr}(\text{adj}(S) \nabla^2 g) \geq 0. \end{aligned} \quad (3.7)$$

All second derivatives of v cancel by equality of mixed partials. Indeed

$$S = \begin{pmatrix} -(v_2)_x & ((v_1)_x - (v_2)_y)/2 \\ ((v_1)_x - (v_2)_y)/2 & (v_1)_y \end{pmatrix}.$$

For a positive-semidefinite 2×2 matrix its adjugate is also positive semidefinite, proving the inequality for convex g . No divergence-free condition is used.

For the piecewise-affine g , convolve with smooth nonnegative mollifiers. The resulting g_δ are convex; their gradients have norm at most $\max_i |a_i|$ and converge to a_i on every open polygon edge. The finitely many vertices have zero arclength measure. Green's formula applies to each convolution, and dominated convergence gives the boundary integral (3.6). A clockwise polygon therefore makes that integral nonpositive.

Now use the actual finite velocities $v_i = b'_i(t_0)$ at $y_i = b_i(t_0)$. Condition (3.1) says precisely that the data $F_i = Jv_i$ are monotone, since

$$(y_j - y_i) \cdot (F_j - F_i) = -\det(y_j - y_i, v_j - v_i) \geq 0.$$

Apply Lemma 3.2 and set $v^{(\ell)} = -JF^{(\ell)}$. Then $\text{sym}(JDv^{(\ell)}) \geq 0$, and $v^{(\ell)}(y_i) \rightarrow v_i$. If P were clockwise, (3.6) would be nonpositive for each of these fields. Passing to the limit in its finite vertex sum would give $D'(t_0) \leq 0$, a contradiction. Thus every long transition polygon is counterclockwise. This argument requires neither convergence of derivatives nor a time-coherent extension of the vertex velocities.

3.5. The local charge

For an ordered collision-free configuration put $\Delta(Y) = \prod_{i < j} (y_j - y_i)$. Join the old ordered configuration at $t_0 - \varepsilon$ to the new ordered configuration at $t_0 + \varepsilon$ by straight interpolation. We compute the limiting increment of $\arg \Delta$.

For a simple counterclockwise k -gon, parameterize its boundary by $\gamma(j+s) = (1-s)v_j + sv_{j+1}$, periodically with period k . For every $0 < d < k$, the loop $h_d(t) = \gamma(t+d) - \gamma(t)$ is nonzero and has winding one. Nonvanishing follows from simplicity, and winding is constant as d varies. For $0 < d < 1$, same-edge chords are constant vectors, while chords passing a vertex interpolate between its consecutive edge vectors. Those vectors cannot be oppositely directed, which would force boundary overlap. Their argument increments are the signed exterior turns; their sum is 2π by the polygon turning theorem, including reflex vertices.

In the limiting transition, particle j moves as $\gamma(j+s)$. For every cyclic gap $d = 1, \dots, k-1$, summing the argument increments over j traces one full h_d , contributing 2π . Each unordered pair is counted twice, with equal argument increments for a vector and its negative. The internal Vandermonde increment is therefore $\pi(k-1)$. Here “fixed” means fixed during the limiting transition, not stationary along the original motion. For each target q outside the cycle, the sum of the moving-to-fixed increments traces $\gamma - q$ once and adds 2π if q is inside, zero otherwise. Pairs both outside the cycle contribute zero. Thus the total is $\pi(k-1+2r)$. Every limiting difference is separated from zero on the compact parameter interval; the actual differences converge uniformly, and their ratios to the limiting ones have logarithms tending uniformly to zero. Their lifted increments converge without discarding multiples of 2π .

For a transposition let $a = a_i - a_j$ and $d(t) = b_i(t) - b_j(t)$. The score order gives $a \cdot d(t_0) = 0$ and $a \cdot d'(t_0) < 0$. Writing $a = cJd$ at t_0 , the derivative inequality is $c \det(d, d') < 0$. By (3.1), this forces $c > 0$ and $\det(d, d') < 0$. The exchanged difference on the bridge is

$$d_\varepsilon(s) = (1-s)d(t_0 - \varepsilon) - sd(t_0 + \varepsilon).$$

Both endpoint dot products with a are strictly positive; hence the whole bridge is in the half-plane $a \cdot d_\varepsilon > 0$. After multiplication by $\bar{a}$, its endpoint arguments tend to $-\pi/2$ and $+\pi/2$. Its lifted argument increment tends to $+\pi$. The limiting exchanged difference itself collides, so it must not be treated by the preceding separation argument. Instead, for a fixed target q group the two factors.

Their limiting product is

$$((1-s)y_i + sy_j - q)((1-s)y_j + sy_i - q).$$

It is nonzero and invariant under $s \mapsto 1-s$, so it retraces itself and has zero increment. Uniform ratio convergence applies to this grouped product. The limiting transposition charge is therefore one. This proves the formula for q_e in (3.3).

3.6. Closing the path and removing source genericity

For fixed A define the strict identity-optimality cone

$$C_A = \left\{ Y : \sum_i a_i \cdot y_i > \sum_i a_i \cdot y_{\rho(i)} \text{ for every } \rho \neq \text{id} \right\}. \quad (3.8)$$

It is convex and contains no collision, since coincident target entries would tie identity with a transposition. The Vandermonde is nonzero there. For generic A , choose a time-circle seam with unique optimum. Reorder the targets by the optimum on each interval, and replace small neighborhoods of all switches by the bridges just described. The resulting path lies in C_A ; its seam endpoints agree with the same labeling. It is a closed loop contractible inside that cone and hence has zero Vandermonde winding.

Reordering multiplies the Vandermonde by a constant sign on each physical interval. As the deleted neighborhoods shrink, the physical increment tends to $-2\pi\mathcal{W}$, and the bridge increments to $\pi \sum_e q_e$. Their sum is zero, proving (3.3) and the event bound. If there is an event, the number of distinct interval optima is no greater than the number of switches around the circle. If there is none, there is one optimum throughout and (3.3) gives $\mathcal{W} = 0$. This proves the bound N , including the zero-winding case.

For arbitrary A , choose a strict witness time for each permutation that is uniquely optimal somewhere. There are finitely many witnesses and competitors, so sufficiently small source perturbations preserve all the strict inequalities. Choose a generic source list in this neighborhood. The targets, their trajectories and $\mathcal{W}$ remain fixed, so its bound transfers to the original distinct count. If there are no strict witnesses, the assertion is immediate.

3.7. Canonical choices and finite open cells

Flatten A into $\mathbb{R}^{2n}$ and use its standard coordinate vectors e_r . Define $\pi_*(t)$ by lexicographically maximizing the coefficients, in increasing powers of λ , of the formal score for

$$A(\lambda) = A + \sum_{r=1}^{2n} \lambda^r e_r. \quad (3.9)$$

Equivalently, maximize $(A \cdot C_\pi(t), C_{\pi,1}(t), \dots, C_{\pi,2n}(t))$ lexicographically. Distinct targets make the vectors $C_\pi(t)$ distinct, so the canonical choice is unique at every time. It always optimizes the original score, the first criterion.

Proposition 3.3 (Canonical finiteness). *Under (3.1), the image of π_* has size at most N . There are at most $2\mathcal{W}$ times at which π_* is not locally constant. An isolated different point value between identical neighboring values is impossible. Thus there are at most N canonical open cells, and their closures cover the original optimal value at every time.*

Proof. At any finite list of times, all canonical choices are simultaneously strict optima for sufficiently small real $\lambda > 0$: each of the finitely many polynomial score differences has positive first

nonzero coefficient. A nearby generic source A' preserves these strict choices. This does not assert that the curve $A(\lambda)$ is itself globally generic. The image is a subset of the finite set of permutations. Choose one witness for each of its values. Their common generic realization proves the image bound, including choices occurring only at a tie point.

Similarly, every finite cyclic list of sampled canonical values has at most $2\mathcal{W}$ adjacent changes: its generic realization requires at least one switch between successive unequal samples. If more than $2\mathcal{W}$ points were points of nonlocal constancy, choose that many plus one distinct points and disjoint small surrounding arcs. Each arc contains two samples with different values, giving a cyclic list with too many changes. The bad set is therefore finite, and the canonical policy is constant on every complementary open arc.

We record the local charge argument needed to exclude isolated spikes. For times $a < b$, put $Y_a = (b_{\pi_*(a)(i)}(a))_i$, define Y_b similarly, and set $L(s) = (1-s)Y_a + sY_b$. A finite generic realization places both endpoints, hence their collision-free chord, in $C_{A'}$. Since Δ^2 is permutation invariant,

$$Q(a, b) = \text{wind}\left(\Delta^2(L) * \text{reverse}(\Delta^2(B|_{[a,b]})), 0\right) = \sum_{a < t_e < b} q_e \geq 0. \quad (3.10)$$

The events on the right belong to the auxiliary generic source A' ; they are not asserted to be the degenerate events of the original source. Its strict endpoint witnesses exclude events at a and b . Indeed, insert the local transition bridges into the reordered physical path. It and L are homotopic relative to their endpoints in the convex cone. Their lifted Vandermonde increments agree; shrinking the bridges gives a difference of $\pi \sum q_e$ from the physical increment. Squaring and dividing by 2π proves (3.10). The left side depends only on the original trajectories and the canonical endpoint choices, not on the auxiliary source. Using one realization for all subdivision endpoints proves additivity. Realizing an arbitrary additional interior sample proves that $Q = 0$ forces the canonical policy to be constant throughout $[a, b]$.

For a sufficiently short interval around any fixed time, each physical pair difference stays in a common open half-plane, as does the straight chord between its endpoint differences. If the canonical endpoint permutations agree, closing with that same labeling gives zero winding for every pair, hence $Q = 0$. An isolated different middle value would contradict the preceding implication. This also proves the converse that a constant canonical policy has zero interval charge, by finite subdivision into such short intervals and additivity.

If the bad set is empty, there is one cell on the whole circle. Otherwise its complementary arcs number at most $2\mathcal{W}$. A boundary value may differ from both neighboring policies; no assertion to the contrary is needed. By continuity, either neighboring original-score optimizer remains optimal at the boundary. Their closed cells therefore cover the original optimal value. This completes the canonical assertion and [Theorem 3.1](#). $\square$

For arbitrary smooth curves this is a qualitative finiteness result, not an effective algorithm for discovering zeros from an unrestricted smooth-function oracle.

3.8. Sufficient fields and a strict extension beyond convex potentials

Suppose the trajectories solve $b'_j = v_t(b_j)$ in an open convex domain U , with $\text{sym}(JDv_t) \geq 0$ throughout U . Integrating along the chord between two targets gives

$$(b_j - b_i) \cdot (Jv_t(b_j) - Jv_t(b_i)) = \int_0^1 d^T \text{sym}(JDv_t(b_i + sd)) d \geq 0,$$

which is (3.1). In particular the Hamiltonian system

$$b'_j = -J\nabla H_t(b_j) \quad (3.11)$$

satisfies the condition when $\nabla^2 H_t \geq 0$ on U ; a positive-definite Hessian makes every pair angle strictly decreasing and every w_{ij} positive. Domain-wide positivity is a sufficient mechanism, whereas finite pairwise velocity data are the actual hypothesis.

Example 3.4 (Breathing square without convex Hamiltonian interpolation). Take $c_1 = (1, 0)$, $c_2 = (0, 1)$, $c_3 = (-1, 0)$, $c_4 = (0, -1)$ and

$$r(t) = 1 + \frac{1}{2} \sin(3t), \quad b_j(t) = r(t)R_{-t}c_j.$$

Since $r \geq 1/2$, these are distinct smooth periodic targets. Every pair difference satisfies

$$\det(b_j - b_i, b'_j - b'_i) = -r(t)^2 |c_j - c_i|^2 < 0, \quad w_{ij} = 1.$$

At $t = 0$, the positions are c_j and the values a Hamiltonian gradient would have to interpolate are $F_j = Jb'_j(0) = c_j + \frac{3}{2}Jc_j$. With cyclic indices, direct calculation gives

$$\sum_{j=1}^4 F_j \cdot (c_{j+1} - c_j) = -4 + \frac{3}{2}4 = 2 > 0.$$

If a differentiable convex H on a convex neighborhood of the square satisfied $\nabla H(c_j) = F_j$, its four supporting-plane inequalities $H(c_{j+1}) \geq H(c_j) + F_j \cdot (c_{j+1} - c_j)$ would sum to $0 \geq 2$. Thus not even these instantaneous velocities admit convex Hamiltonian interpolation. The pairwise theorem nevertheless applies. This uniform breathing is a strict scope witness, not a new sharpness construction; its positive common scaling preserves the rotational matching orders.

Corollary 3.5 (Full-twist braid class). *If all $w_{ij} = 1$, the labeled motion represents the central clockwise full twist in the pure braid group.*

Proof. Choose a line direction that is a regular value of all pair-direction maps modulo π , avoiding the initial directions. Each monotone map turns through 2π , so projection onto the perpendicular axis has exactly two transverse crossings per pair, all with clockwise handedness. A small periodic C^1 perturbation separates simultaneous crossings without changing their signs or counts: derivatives are nonzero near each crossing, projected differences stay away from zero elsewhere, and compactness supplies positive collision separation. The braid class is unchanged. Casson's characterization, reported by Elrifai and Morton [21, Section 5], says that a positive braid with exactly two crossings per pair is Δ_n^2 . This is the clockwise full twist when clockwise crossings are called positive, and Δ_n^{-2} under the opposite convention. It is stronger than a statement about pairwise linking numbers alone, but is an application of the classical characterization, not a new braid classification. $\square$

3.9. Rotation and nonlinear sharp examples

Theorem 3.6 (Sharp rotational count). *For arbitrary planar lists of size $n \geq 2$, at most $n(n-1)$ different permutations uniquely minimize $\sum_i |a_i - R_{-t}b_{\pi(i)}|^2$ for some t . The bound is attained for every n , and for $n \geq 3$ persists on a nonempty open set with both lists noncollinear. For $n = 1$ the count is one.*

Proof. For distinct targets use $H(x) = |x|^2/2$; every pair winds once clockwise, so Theorem 3.1 gives $2^*W = n(n-1)$ for every A . If target labels coincide, interchanging those columns leaves every score unchanged, so no permutation is unique and the upper bound is immediate.

For sharpness take, with indices from 0 to $n-1$, $a_i = (i, 0)$ and $b_j = (2^j, 4^j)$. The strict rearrangement inequality matches the increasing projections of $R_{-t}b_j$ to increasing i . Pair slopes

are $2^p + 2^q$ for $p < q$, all distinct by their two-adic valuations. Each pair has two projection ties, and none of these times coincide. Thus there are $n(n-1)$ strict projection-order sectors. A specified strict order occupies at most one sector: each pair inequality defines an open semicircle, and a nonempty intersection is connected after cutting at the antipode of one of its points. Thus the optima in the sectors are distinct. Choose one strict witness in each sector. Their positive gaps persist on an open neighborhood of the input, which contains noncollinear lists when $n \geq 3$. The upper bound precludes any additional distinct optima. Generic sources can also be chosen there. $\square$

Corollary 3.7 (Polynomial shear conjugacies). *Let p be any fixed real polynomial, $|c_j| \leq 1$ with distinct c_j , and*

$$F_\varepsilon(u, v) = (u, v + \varepsilon p(u)), \quad H_\varepsilon(x, y) = \frac{1}{2}[x^2 + (y - \varepsilon p(x))^2], \quad b_j(t) = F_\varepsilon(R_{-t}c_j). \quad (3.12)$$

For sufficiently small nonzero $|\varepsilon|$, at most $n(n-1)$ permutations are uniquely optimal, for every source list. The count is independent of $\deg p$, though the permitted amplitude can depend on p . For every $n \geq 2$ there are genuinely nonrigid examples attaining the bound.

Proof. The shear is a symplectic diffeomorphism, and direct differentiation gives $b'_j = -J\nabla H_\varepsilon(b_j)$. Its Hessian satisfies

$$(H_\varepsilon)_{yy} = 1, \quad \det \nabla^2 H_\varepsilon = 1 - \varepsilon(y - \varepsilon p(x))p''(x).$$

On a fixed box containing the unit disk, small enough amplitude keeps this determinant positive and all trajectories in its interior. All $F_{s\varepsilon}$, $0 \leq s \leq 1$, are injective, so every pair-difference loop is homotopic through nonzero loops to $R_{-t}(c_j - c_i)$. Hence $w_{ij} = 1$.

For a concrete nonlinear case take $p(u) = u^2$, $\varepsilon = 1/10$. On $[-2, 2]^2$ the determinant is at least $3/5$, while trajectories from the unit disk satisfy $|x| \leq 1$, $|y| \leq 11/10$. For $c_1 = 0$ and $c_2 = (r, 0)$, the squared separation is $r^2 + \varepsilon^2 r^4$ at time zero and r^2 at $\pi/2$, so the motion is not rigid. Generally its second harmonic is permutation-dependent when source vertical coordinates vary.

Rescale a sharp rotational target set into the unit disk and keep its finitely many strict witnesses. For sufficiently small nonzero quadratic shear they persist. The upper bound then gives equality. To ensure actual deformation, choose a generic nearby target set for which a pair has different squared separations at two fixed times; this is an open condition and its failure is a proper polynomial constraint (the displayed two-point pair proves properness). Such choices coexist with the open sharp-witness condition. The amplitude may depend on the instance; no fixed-amplitude all-size saturation is claimed. $\square$

4. A charge certificate and exact kinetic enumeration

A bound on matching changes does not bound failures of a solver's internal certificates. We instead compute the nonnegative geometric charge of an entire interval. The coordinate-lexicographic source selection in (3.9) makes this certificate available without any promise of unique or transverse optima for the given sources. Throughout this section write $\mathcal{B} = 2\mathcal{W}$ for the raw charge budget, so that the representative bound in (3.2) is $N = \max(1, \mathcal{B})$.

Theorem 4.1 (Canonical interval charge). *Let the smooth, collision-free periodic targets satisfy (3.1), and let A be arbitrary. Write $\pi_*(t)$ for the coordinate-lexicographic source policy. For $a < b$, put*

$$Y_a = (b_{\pi_*(a)(i)}(a))_i, \quad Y_b = (b_{\pi_*(b)(i)}(b))_i, \quad L(s) = (1-s)Y_a + sY_b.$$

The chord L is collision-free. With $Z_B(t) = \Delta(B(t))^2$ and $Z_L(s) = \Delta(L(s))^2$, define

$$Q(a, b) = \text{wind}(Z_L * \text{reverse}(Z_B|_{[a,b]}), 0). \quad (4.1)$$

Then $Q(a, b)$ is a nonnegative integer, is additive under subdivision, and vanishes exactly when π_* is constant on $[a, b]$. Different endpoint policies imply positive charge. Over a full period the charge is $\mathcal{B}$. These assertions include repeated sources and persistent original-score ties.

Proof. For any finite list of times, sufficiently small positive values of the formal source perturbation in (3.9) make the selected policies simultaneously strict optima. A nearby generic source list A' preserves these strict inequalities. This is an existence argument; neither perturbation is an algorithmic subroutine.

For the two endpoints, Y_a, Y_b lie in the same strict identity-optimality cone $C_{A'}$ of (3.8). Its convexity and absence of collisions prove the chord assertion. Squared Vandermonde values are permutation-invariant, so the paths in (4.1) join exactly. The generic bridge identity in Theorem 3.1, applied to A' , expresses this winding as the sum of its positive event charges between the endpoints. This proves nonnegativity and positivity when the endpoint policies differ. A common finite realization of three endpoint policies puts their chords in one convex collision-free cone; the argument increments telescope, proving additivity.

If $Q(a, b) = 0$, include any interior query time in the finite realization. The generic source has no transition between its endpoint witnesses, so the three selected policies agree. Conversely, for a constant canonical policy, partition the interval finely enough that every physical pair difference and its same-label closing chord stay in one open half-plane. Each pair loop then has zero winding, and additivity gives $Q(a, b) = 0$. Over a full period the endpoint configurations agree and the chord is constant. The reversed physical squared Vandermonde has winding $2\mathcal{W}$, proving the total-charge assertion. $\square$

Charge is not an event count: a single event may consume several units, and a long positive-charge interval may have equal endpoint policies. Proposition 3.3 bounds canonical bad times by $\mathcal{B}$ and open-cell representatives by N . A boundary's point policy may differ from both adjacent policies; it is not required in the output below.

4.1. Exact charge and point-germ oracles

In a rational chart, let all coordinates have a common positive denominator. At a rational sample the canonical assignment maximizes a vector of $2n + 1$ rational coefficients: original correlation first, then the successive source-coordinate perturbation coefficients. Distinct permutations give different vectors because targets are distinct. Assignment arithmetic with lexicographic comparisons is exact. Alternatively, clear denominators in all entry vectors, bound their integer magnitudes by $M_0 \geq 1$, and encode a k -component vector using radix $B_0 = 2nM_0 + 2$. In two assignment totals, the first different component dominates all later radix terms. Ordinary exact assignment optimization implements the vector oracle with polynomial bit length.

At a rational sample z_0 , also compute the left and right germ policies. Within each source coefficient expand its complete numerator Taylor polynomial, multiplying Taylor coefficient r by $(-1)^r$ for the left germ. Order by source-perturbation degree first, Taylor degree second. This is the order $0 < \lambda \ll |z - z_0| \ll 1$: source selection is made at a punctured time before taking its one-sided limit. A sample is regular precisely when its point and two germ policies coincide. Coordinate numerator degree at most $2D$ gives $O(nD)$ rational criteria. A positive common denominator does not affect comparisons.

For (4.1), choose a rational ray avoiding endpoint values. Isolate the real zeros of each rotated imaginary numerator and determine the real-part sign at every root. On the positive ray an upward

crossing contributes +1, a downward crossing −1, and an even-order tangency zero. Exact root isolation and algebraic sign determination have polynomial bit complexity [15]; repeated roots are handled by order and sign. A gcd checks common real numerator roots.

It suffices to work with squared pair factors, rather than a Vandermonde polynomial of unnecessarily large degree:

$$Q(a, b) = \sum_{i < j} \text{cross}((L_j - L_i)^2) - \sum_{i < j} \text{cross}((b_j - b_i)^2|_{[a, b]}). \quad (4.2)$$

At either join the unordered multisets of squared pair differences agree, so principal-argument endpoint terms cancel. Among $2\binom{n}{2} + 1$ integer ray slopes, one avoids the at most $2\binom{n}{2}$ endpoint directions. Dropping positive denominators changes no argument increment; positive radial joins can restore endpoint moduli without contributing winding. The endpoint avoidance also ensures that none of the rotated imaginary numerators is identically zero. Thus the crossing procedure has only finitely many roots to inspect, even when some physical pair angle is constant on the entire interval.

Theorem 4.2 (Checked rational-Fourier registration). *Let the sources be arbitrary rational planar points, allowing repetitions. Let each target coordinate be a densely encoded rational trigonometric polynomial of degree at most D , with $D \geq 1$ by convention and period 2π . A deterministic polynomial-bit-time algorithm checks that the targets never collide and that (3.1) holds throughout the period. On passing these checks it computes every pair winding and*

$$\mathcal{B} = 2\mathcal{W} \leq n(n-1)D.$$

It then outputs every boundary between canonical open cells, the adjacent cell policies, and at most $N = \max(1, \mathcal{B})$ distinct representatives whose closed cells cover the original optimal matching value at every phase. It also computes the exact minimum squared matching distance over phase, optionally optimizing a common translation, with a matching witness and optimal phase represented by algebraic sine and cosine. These phase coordinates, the value and any translation have algebraic degree $O(D)$.

No genericity, Hamiltonian-flow, or convex-extension promise is required. The output need not list all tied permutations, all optimal faces, or every canonical policy occurring only at a boundary. If $\mathcal{B} = 0$, there is one constant canonical policy. For $n = 1$ only univariate phase optimization is needed.

Each boundary is represented by its chart index, a rational polynomial, and an isolating interval selecting one real root. Cell endpoints are ordered within their charts, whose seams are chosen to be regular. The reported adjacent policies are therefore enough to recover the cyclic open-cell decomposition; repeated occurrences of one permutation need not be duplicated in the separate representative list. The optimization output consists of one minimizing phase and one realizing permutation, not a description of the complete set of optimal phases.

We prove the algorithmic claims in the following subsections. Polynomial time refers to the dense Fourier encoding, including D and coefficient bit length, not a sparse encoding with arbitrarily large binary harmonic indices. This theorem concerns Fourier paths; no effective algorithm for an arbitrary smooth path is asserted.

4.2. Charts and checked input geometry

For a rational unit center $(\cos \phi, \sin \phi)$ put $t = \phi + 2 \arctan z$, $-1 \leq z \leq 1$. Complex powers of $(1 - z^2) + 2iz$, followed by the rational center rotation in each harmonic, give coordinate numerators

of degree at most $2D$ over the denominator

$$q_D(z) = (1 + z^2)^D > 0.$$

Two antipodal centers give consecutive half-circle charts. Their endpoint values agree because they represent the same periodic Fourier functions. Coefficient lengths remain polynomial in the dense input length.

For each pair write $d = P/q_D$, with $P = (P_x, P_y)$. Noncollision means that P_x, P_y have no common real root on either closed chart; an identically zero pair is rejected. A gcd and exact root isolation check this, including endpoints. Moreover,

$$\det(d, d'_z) = \frac{\det(P, P')}{q_D^2}.$$

Since $dt/dz > 0$, the required time-sign condition is exactly nonpositivity of this numerator. Isolate its roots, test one rational point in each complementary interval, and test the endpoints. Identically zero determinants are allowed. A common ray avoiding all pair values at the seams gives the signed crossing sum of each *unsquared* pair difference over both charts. Its negative is the clockwise integer w_{ij} .

The checked angular condition gives $w_{ij} \geq 0$. A generic line through the origin intersects a nonzero pair-difference trigonometric curve at most $2D$ times: its scalar line equation is a nonzero trigonometric polynomial of degree at most D . Each monotone full turn uses two crossings, so $w_{ij} \leq D$ and $\mathcal{B} \leq n(n-1)D$. Zero winding causes no exception to the checks.

4.3. Regular samples and effective small intervals

There are at most $\mathcal{B}$ bad sample times by Proposition 3.3. Test $2\mathcal{B} + 1$ distinct rational-circle directions until both a direction and its antipode are regular; at most $2\mathcal{B}$ candidates are forbidden. Place the two chart centers a quarter turn from these seams. For $\mathcal{B} = 0$ the first candidate suffices. Inside a positive-charge interval, $\mathcal{B}+1$ distinct rational candidates in its middle third likewise supply a regular balanced split. Child lengths are at most $2/3$ of the parent length, and each depth adds only $O(\log(\mathcal{B} + 1))$ bits to sampling denominators when $\mathcal{B} > 0$.

We need an effective threshold below which equal endpoint policies force zero charge. For each pair, minimize $|d|^2$ exactly over the two charts by stationary-root isolation, including endpoints and constant branches. Noncollision makes its algebraic minimum positive. Refine its isolating interval until it has a positive rational lower endpoint ℓ_{ij} . Set

$$d_0 = \min(1, \min_{i < j} \ell_{ij}) > 0.$$

Then $|d| \geq d_0$: below one a squared distance is no greater than its distance, and above one the bound one suffices. For an empty pair set take $d_0 = 1$. On $[-1, 1]$ the coefficient 1-norms of

$$P'_x q_D - P_x q'_D, \quad P'_y q_D - P_y q'_D$$

bound the corresponding derivative numerators. Since $q_D \geq 1$, their sum bounds $|d'_z|$. Let $M_1 \geq 1$ bound these sums for all charts and pairs. A valid rational threshold is

$$\eta = \frac{d_0}{4M_1}. \tag{4.3}$$

On a shorter interval the physical pair path and its same-label endpoint chord stay within distance $d_0/4$ of the initial pair vector, hence in one open half-plane excluding zero. Equal canonical endpoint labels therefore imply zero charge by (4.2).

These minima are values of rational functions of degree $O(D)$ at algebraic points of degree $O(D)$. Resultant height bounds and separation from zero give polynomial description lengths and polynomially many refinements to obtain the positive ℓ_{ij} . Thus (4.3) is computed from the input, not supplied as an unquantified condition-number promise.

4.4. Uniform separation without enumerating competitors

Include all $2n+1$ source-score coefficient matrices, not just original correlation. Clear one common denominator from their entry polynomials on both charts and let $M_0 \geq 1$ bound the resulting integer coefficient magnitudes. Every nonzero coefficient polynomial of a formal permutation-score difference has degree at most $2D$ and height at most $2nM_0 \leq 2^h$. This includes cases where the primary difference is identically zero.

The product of any two such nonzero polynomials has degree at most $m = 4D$ and height at most 2^b , where

$$b = 2h + \lceil \log_2(2D + 1) \rceil + 1.$$

These bounds do not require generating the factorial competitor family. Its primitive square-free part S is an integer factor with height at most $2^m \sqrt{m+1} 2^b \leq 2^k$, for $k = b + 2m + 2$. Indeed, factor coefficients are bounded by 2^m times Mahler measure; an integer complementary factor has measure at least one, and the measure of a polynomial is at most its Euclidean coefficient norm. Cauchy's bound places the roots of S within modulus 2^{k+1} . Its nonzero integer discriminant has absolute value at least one. In

$$|\text{disc } S| = |\text{lead } S|^{2r-2} \prod_{i < j} |\alpha_i - \alpha_j|^2,$$

bound all differences except a selected one by 2^{k+2} . This yields the conservative separation

$$|\alpha_i - \alpha_j| > \delta, \quad \delta = 2^{-\lceil m^2(k+2)+2 \rceil}. \quad (4.4)$$

It applies to distinct real roots from any two members of the full coefficient-difference family, including a polynomial paired with itself. The exponent has polynomial size despite the ungenerated family size.

4.5. Complete subdivision and global minimization

Compute canonical endpoint policies and charge on each chart. Discard every zero-charge interval, which has constant policy by [Theorem 4.1](#). Split positive-charge intervals at regular balanced samples until their widths are below $\min(\eta, \delta)$. A terminal interval has different endpoint policies by (4.3), and at most one distinct candidate coefficient root by (4.4). The earliest source coefficient whose endpoint-policy difference is not identically zero has opposite signs on the two regular endpoint neighborhoods. Its unique interior root is therefore the canonical boundary. Isolate it and output the two adjacent policies. A regular endpoint root cannot replace this interior root: constancy on the endpoint neighborhoods would still require an interior sign change.

At each depth at most $\mathcal{B}$ intervals have positive charge: they are disjoint and their positive integer charges sum to $\mathcal{B}$. Balanced subdivision has depth $O(1 + \log(1/\eta) + \log(1/\delta))$, polynomial in the input length. Including discarded children, the number of nodes is polynomial, and each uses polynomially many assignment and univariate-algebra operations of polynomial degree and bit size. This proves termination and cell enumeration. Certified earlier stops, such as a unit-charge interval with one endpoint-difference root, can shorten subdivision but are not needed for the bound. A terminal charge greater than one is allowed. For $\mathcal{B} = 0$, both initial charts have zero charge and no subdivision is necessary.

Let Π be the at most N distinct open-cell policies. Each original score is optimal on its cell, since original correlation is the first source-lexicographic criterion. At a boundary, a neighboring representative remains originally optimal by continuity. Thus the family covers the original optimum at every phase, including persistent and point-only ties. For branch energies f_π ,

$$\min_t \min_\pi f_\pi(t) = \min_{\pi \in \Pi} \min_t f_\pi(t).$$

This does not enumerate all tied permutations or the root's possibly different canonical point policy. If desired, that policy can be computed by lexicographic assignment over the isolated root's real algebraic field; it is unnecessary for value coverage.

For free translation use the convention $\sum_i |a_i + \tau - b_{\pi(i)}(t)|^2$. Its unique optimal translation is $\tau(t) = \bar{b}(t) - \bar{a}$; equivalently, center both lists in the squared-distance energy. The correlation change is permutation-independent, so the same Π suffices. This convention agrees with the Fourier implementation; the rigid appendix states separately its convention of transforming the targets. Each full branch energy, including its possibly nonconstant target-norm term, has numerator degree at most $4D$ over $(1 + z^2)^{2D}$. After cancelling the common denominator factor, its derivative numerator has degree at most $4D + 1$. Isolate its real roots, include endpoints, and compare algebraic values exactly. Treat constant branches and equal values explicitly. Before forming a value resultant, remove common complex denominator factors from the root polynomial to avoid a spurious zero resultant. The selected half-angle coordinate, its phase sine and cosine, the value and the translation belong to a field of degree $O(D)$. The real angle itself need not be algebraic. These steps also handle $n = 1$ and finish [Theorem 4.2](#). No degree-independent nonlinear running-time estimate is claimed.

5. Necessity and dimensional limits

5.1. Fixed winding can coexist with factorial diversity

Proposition 5.1. *For every $n \geq 4$ there are fixed distinct planar sources and smooth collision-free periodic labeled target paths, generated by one common smooth time-periodic Hamiltonian with compact spatial support, such that every pair winds once clockwise and every one of the $n!$ permutations is uniquely optimal at some time.*

Proof. Start the ordered target configuration at the distinct source configuration A . For every permutation σ , let $B_j^\sigma = a_{\sigma(j)}$. At this configuration σ^{-1} is the unique zero-cost matching. The ordered configuration space of distinct planar points is path connected: move points one at a time to separate staging positions and then to their destinations, avoiding the finitely many stationary points at each step. Smooth the paths and make all derivatives vanish at their endpoints. Choose such a path from A to each B^σ , traverse it and its exact reverse, and concatenate all these excursions. Each pair-difference path retraces itself, so each excursion has zero pair winding. Append a clockwise rigid turn of A , with a smooth angle parametrization flat at the endpoints. The complete loop is smooth and periodic, realizes all the strict witnesses, and has unit clockwise winding for every pair.

To realize these paths by one Hamiltonian, write $u_j = b'_j$ and choose a smooth radial bump χ , equal to one near zero, supported in a disk of radius less than one third of the minimum pair separation. Set

$$H(t, x) = \sum_j \chi(x - b_j(t)) (Ju_j(t)) \cdot (x - b_j(t)).$$

The supports about different particles are disjoint, so

$$\nabla_x H(t, b_j(t)) = Ju_j(t), \quad -J\nabla_x H(t, b_j(t)) = b'_j(t).$$

Smooth periodicity and compact spatial support follow from the construction. No convexity is claimed. $\square$

The excursions reverse relative angles; thus this does not contradict [Theorem 3.1](#). It rules out replacing the pointwise sign condition by final integer winding alone, even within smooth Hamiltonian dynamics. It is not a lower bound for bounded-degree Fourier input, and the elementary path-realization construction is not asserted to be a new isotopy theorem.

5.2. Thirteen matchings for four points in three dimensions

For fixed-axis source rotation

$$R_t(x, y, z) = (x \cos t - y \sin t, x \sin t + y \cos t, z),$$

the permutation-dependent score has the form

$$S_\pi(t) = \alpha_\pi \cos t + \beta_\pi \sin t + \gamma_\pi,$$

where

$$\begin{aligned} \alpha_\pi &= \sum_i (a_{ix} b_{\pi(i)x} + a_{iy} b_{\pi(i)y}), \\ \beta_\pi &= \sum_i (-a_{iy} b_{\pi(i)x} + a_{ix} b_{\pi(i)y}), \\ \gamma_\pi &= \sum_i a_{iz} b_{\pi(i)z}. \end{aligned}$$

The axial constant term is a genuine change from the planar correlation polygon.

Proposition 5.2. *There are rational four-point lists in $\mathbb{R}^3$ with at least thirteen distinct uniquely optimal squared-distance matchings during one fixed-axis turn. Both lists can affinely span $\mathbb{R}^3$, and the separation persists after optimizing a common translation and under small input perturbations.*

Proof. Use the rows of

$$A = \begin{pmatrix} -3 & 0 & -2 \\ -1 & 0 & 2 \\ 1 & 0 & 2 \\ 3 & 0 & 0 \end{pmatrix}, \quad B = \begin{pmatrix} -20 & -6 & -5 \\ 40 & -8 & -1 \\ -20 & -10 & -2 \\ -40 & -2 & 3 \end{pmatrix}.$$

Each row of [Table 1](#) gives a different permutation and a rational unit rotation $(\cos t, \sin t) = (c/d, s/d)$. Direct integer substitution gives

$$m = \min_{\tau \neq \pi} [(\alpha_\pi - \alpha_\tau)c + (\beta_\pi - \beta_\tau)s + (\gamma_\pi - \gamma_\tau)d] > 0.$$

The table therefore certifies strict score gaps m/d against every competitor; the squared-cost gaps are twice as large. Translation elimination subtracts $n|R_t \bar{a} - \bar{b}|^2$ from every squared cost, so it preserves all gaps.

The minimum displayed score gap is $40/61$. Change only the first source to $(-3, 1/200, -2)$. For its targets j, k , the change in a competing score difference is

$$\frac{1}{200} [-(b_{jx} - b_{kx}) \sin t + (b_{jy} - b_{ky}) \cos t].$$

Table 1: Exact witnesses for [Proposition 5.2](#). Labels start at one; $c^2 + s^2 = d^2$ in every row.

π	c	s	d	m
(4, 3, 1, 2)	153	104	185	200
(1, 4, 3, 2)	3	-4	5	8
(3, 4, 1, 2)	3	4	5	4
(1, 3, 4, 2)	21	220	221	892
(3, 2, 4, 1)	0	1	1	8
(2, 3, 4, 1)	-11	60	61	56
(2, 3, 1, 4)	-255	32	257	256
(2, 1, 3, 4)	-3	-4	5	6
(2, 1, 4, 3)	-7	-24	25	88
(2, 4, 1, 3)	-11	-60	61	40
(1, 2, 4, 3)	-15	-112	113	160
(1, 4, 2, 3)	0	-1	1	6
(4, 1, 3, 2)	323	-36	325	288

The coordinate spreads of B are 80 and 8, so its absolute value is at most $88/200 = 11/25$. Every displayed gap therefore remains at least $329/1525 > 0$. The last three sources are noncollinear in the xz -plane and the first now lies off that plane. The determinant of $(b_2 - b_1, b_3 - b_1, b_4 - b_1)$ is -2840 . Thus both lists are full-dimensional. The finite strict inequalities persist under perturbation. $\square$

Among $n \geq 2$, four is the smallest possible cardinality for exceeding $n(n-1)$, because $n! = n(n-1)$ for $n = 2, 3$. Before perturbation all $a_{iy} = 0$, and the score matrix is

$$C(t) = x(u \cos t + v \sin t)^\top + zw^\top,$$

with $A_i = (x_i, 0, z_i)$ and $B_j = (u_j, v_j, w_j)$. Its rank is at most two at every angle. Thus pointwise rank two alone does not imply the planar bound when a rank-one constant term is admitted. We claim neither an exact total of thirteen nor an asymptotic lower bound in three dimensions.

There is a limited upper bound when only r source heights differ from one common value. Put $m = n - r$ and $H(0) = H(1) = 1$, $H(m) = m(m-1)$ for $m \geq 2$. Fixing the injection of the r exceptional sources leaves a planar rotational problem on the other m sources: their common height times the sum of the remaining target heights is independent of their bijection. There are $n!/(n-r)!$ such injections. A globally unique matching must also be unique in that restriction. Consequently the number is at most

$$\min\{n!, n!H(n-r)/(n-r)!\}.$$

This is $O(n^{r+2})$ for fixed r , not a uniform polynomial bound for unrestricted heights.

6. Reproducibility, scope, and remaining questions

The source package includes portable Python exact-arithmetic code using only the standard library. Its Fourier entry point constructs rational charts, checks noncollision and the angular sign condition, computes pair windings, selects regular seams, isolates canonical changes, and minimizes the original squared-distance objective, optionally eliminating translation. It outputs polynomial/root-isolator representations and charge certificates. The structural theorems are mathematical dependencies, not machine-verified conclusions of the software. The specialized support-polygon algorithm in [Appendix A](#) and the common-metric quartic algorithm in [Appendix B](#) are constructive

mathematical algorithms; separate implementations of those two algorithms are not included. In particular, the Fourier program does not accept an anisotropic residual metric.

The bounded end-to-end suite has eighteen geometry tests and twelve controller tests. It includes stationary configurations, angular stalls, three turns, nonlinear deformation, repeated sources, all-zero sources, translation, and breathing motion violating convex-Hamiltonian interpolation. For ten inputs of size at most three, a separate factorial reference enumerates coefficient roots and open-cell policies without an assignment solver or charge oracle. All eleven successful solve inputs, including a four-point breathing square, are compared against optimization over every permutation. These checks share the algebraic kernel, and the objective comparisons share the phase optimizer; they do not independently validate those components. Invalid input guards are tested separately.

The three-dimensional witness uses exact integer comparisons at thirteen rational rotations, plus direct rational squared-distance and translation checks. The transport witness includes its exact dual potentials and positive off-support slacks. Discovery searches are not used by the certificate checks. A separate weighted-path check enumerates 46 hull-terminal and 18 interior-terminal cases; it checks shifted disks and signed face potentials, including a four-point equality example. It is not an all-size or multi-defect flow verification. A deterministic parabolic-fan check generates the sharpness construction for $N = 2, \dots, 7$, comparing all 415 simple paths at the strict witnesses and switch parameters, and checking 2884 directed cycles at both certified interval endpoints. The all-size sharpness theorem rests on its written proof, not that finite check. The real-amplitude extension is proved analytically; no exhaustive real-margin flow solver is supplied. The algorithms use no floating-point tolerance for a mathematical decision. Conservative worst-case root-separation thresholds may be costly; optional sufficient unit-charge and polynomial-dual early exits improve the small examples but do not strengthen the general complexity theorem. No large-degree performance claim or independent formal verification is made. The polynomial-bit-time guarantee follows from the bounds on the assignment oracles, subdivision depth, polynomial degrees, coefficient heights, and exact root separation in [Section 4](#); finite test success is not evidence for that asymptotic guarantee. The optional `--max-nodes` switch is an explicit user limit: exhausting it raises an incomplete-run error and never returns a complete solution.

The output contract matters. Canonical open-cell representatives cover the original optimal value at every time; the implementation does not enumerate all tied permutations, all optimal-face changes, or every point-only canonical selection. Arbitrary smooth motion has a qualitative count theorem, whereas the effective theorem requires the dense rational Fourier encoding. Nonmonotone angular motion, collisions, unequal-cardinality matching, and heterogeneous residual metrics do not inherit that theorem. The appendices give only the explicitly stated specialized weighted extensions.

The counterexamples leave substantial mathematical questions. What is the correct growth rate for fixed-axis three-dimensional rotation? Does arbitrary-mass planar rotational transportation admit a uniform polynomial support-size bound? Can broader trajectory conditions retain a positive assignment-switch certificate? The present results do not settle these questions. The targeted literature comparison in [Section 2](#) also does not establish complete priority clearance or external mathematical validation.

A. Rigid registration with degenerate inputs

Rigid rotation admits a more direct, degeneracy-safe support-polygon algorithm with a sharper arithmetic-operation bound than the general Fourier construction. Write centered complex coordi-

nates $a_i^0 = a_i - \bar{a}$ and $b_j^0 = b_j - \bar{b}$, and put

$$z_\pi = \sum_i \overline{a_i^0} b_{\pi(i)}^0, \quad Q = \text{conv}\{z_\pi : \pi \in S_n\}, \quad C = \sum_i |a_i^0|^2 + \sum_j |b_j^0|^2. \quad (\text{A.1})$$

For a unit complex multiplier ρ acting on targets, the optimal translation is $\tau = \bar{a} - \rho \bar{b}$, independently of the permutation. The centered energy is $C - 2 \text{Re}(\rho z_\pi)$. Hence the global minimum is

$$C - 2 \max_{v \in \text{vert } Q} |v|. \quad (\text{A.2})$$

Maximizing a norm over a finite convex hull requires only its vertices, by the triangle inequality.

Lemma A.1 (Distinct vertices under degeneracy). *For all planar input lists with $n \geq 2$, the number V of distinct vertices of Q is at most $n(n-1)$, even when each vertex has several realizing permutations.*

Proof. For each vertex v_k , choose a direction strictly exposing it, and let $E_k = \{\pi : z_\pi = v_k\}$. These nonempty classes are disjoint. A small input perturbation preserves the strict gap, in each chosen direction, between members of E_k and all other permutations. For a fixed nonzero direction d and distinct permutations π, σ , the score-equality condition $d \cdot (z_\pi - z_\sigma) = 0$ is a nonzero real polynomial in the input coordinates. To see this, choose a row i where their target labels differ, set $a_i = 1$ and other source rows zero, and set $b_{\pi(i)} = d$ and other targets zero. The score difference is $|d|^2 > 0$. Centering subtracts a common correlation and changes no difference. A finite union of these proper polynomial zero sets has empty interior. Thus a perturbation can make each selected direction have a unique winner still in E_k . These are V distinct unique permutations, so [Theorem 3.6](#) gives the bound. The argument is existential; the algorithm does not perturb the data. $\square$

Theorem A.2 (Exact rigid registration). *For rational planar lists with $n \geq 2$, the entire correlation polygon can be enumerated with at most $2V$ exact assignment queries, where $V \leq n(n-1)$; a point uses two calls and a segment four. One global orientation-preserving registration and a realizing bijection are computable with $O(n^5)$ integer arithmetic operations and deterministic polynomial bit complexity. Exact output needs at most one square root. A second branch handles orientation-reversing motions.*

Proof. Clear rational denominators so all pair correlation vectors C_{ij} are integral, and put $Z_\pi = \sum_i C_{i, \pi(i)}$. For an integer direction $d \neq 0$, take $t = (-d_y, d_x)$ and maximize $(d \cdot Z_\pi, t \cdot Z_\pi)$ lexicographically. This returns a vertex, since tangent maximization selects an endpoint of the first exposed face. It need not select a unique permutation representing it. Set $B_t = n \max_{ij} |t \cdot C_{ij}|$ and $M = 2B_t + 1$. One assignment solve with weights

$$w_{ij} = M(d \cdot C_{ij}) + t \cdot C_{ij}$$

implements the lexicographic objective exactly: an integer primary gain of one outweighs every possible secondary loss.

Call this oracle at $d = (-1, 0), (1, 0)$ to get the lexicographic extremes p, q . If they agree the polygon is a point. Otherwise explore the two oriented chords (p, q) and (q, p) . For a chord (p, q) use the right normal $d = (q_y - p_y, p_x - q_x)$ and query a vertex r . If $d \cdot r = d \cdot p$, emit the supporting edge $p \rightarrow q$. If the inequality is strict, recurse on (p, r) and (r, q) . The query value cannot be smaller than that of p , a feasible point.

For a two-dimensional convex polygon, intermediate vertices of the counterclockwise chain from p to q are strictly on the right side of its chord. A strict query therefore returns an intermediate vertex of that chain and splits it into disjoint subchains. A non-strict query certifies the

corresponding edge. The two roots cover the boundary; each strict query discovers a new vertex. There are $V - 2$ strict and V terminal calls, plus two initialization calls, totaling $2V$. For a segment both roots terminate, giving four calls; the two-dimensional chain argument is not used in that case. This is the classical dichotomic support-enumeration construction [6, 7], applied to the bounded structured polygon.

For bit complexity, let L be total rational input length and let D_0 be a product of the input denominators. Write

$$\alpha_i = D_0 a_i, \quad \beta_j = D_0 b_j, \quad u_i = n\alpha_i - \sum_k \alpha_k, \quad v_j = n\beta_j - \sum_k \beta_k.$$

Then $Z_\pi = \sum_i \bar{u}_i v_{\pi(i)} = n^2 D_0^2 z_\pi$. The coordinate bit lengths are $O(L + \log n)$. Oracle directions are normals to differences of original polygon vertices, not successively refined numerical angles; their bit lengths and the lexicographic weight bit lengths remain polynomially bounded. With a classical $O(n^3)$ exact Hungarian implementation, at most $2n(n-1)$ queries give $O(n^5)$ arithmetic operations on polynomial-sized integers. Denominator clearing and all comparisons have polynomial bit cost.

Compare squared vertex norms rationally to select a winner v . For $v \neq 0$, set $\rho = \bar{v}/|v|$ and $\tau = \bar{a} - \rho \bar{b}$; if $v = 0$ choose $\rho = 1$. This proves the exact output assertion. For reflections repeat with the conjugated target list and choose the better result. No exact transcendental angle representation is required. $\square$

For $n = 1$ translation fits the pair exactly at every angle. The algorithm in [Theorem A.2](#) retains one realizing permutation per vertex; it does not list every tied permutation, a potentially factorial output. The supplied implementation uses the more general Fourier controller, not this specialized support-polygon algorithm.

Corollary A.3 (Rigid angular envelope). *After optimizing translation and matching, the rigid angular cost is constant, or every local minimum is strict and their number in one full turn is at most $V \leq n(n-1)$.*

Proof. With $u(\theta) = (\cos \theta, \sin \theta)$ the cost is $E(\theta) = C - 2h_Q(u(\theta))$. On a vertex normal interval the support function is one sinusoid; for a nonzero vertex it has at most one local maximum on the whole circle. At a true support switch from p to q ,

$$h'_+(\theta) - h'_-(\theta) = (-\sin \theta, \cos \theta) \cdot (q - p) = |q - p| > 0.$$

Thus a switch cannot be a local minimum of E . The same sign applies to the two switches of a segment. The average of all centered correlations is zero, since each target occurs equally often in each row. It is a positive combination of all generating points, so $0 \in \text{relint } Q$. A singleton polygon is therefore $\{0\}$ and gives constant cost. Otherwise no vertex is zero; no active branch is constant, and its local minima are isolated and strict. Summing over the vertices proves the claim. $\square$

For both orientations, the number of distinct uniquely optimal permutations is at most $\min(n!, 2n(n-1))$. Apply [Theorem 3.6](#) separately to the two branches. All-size sharpness of that union is not asserted.

B. A common fixed anisotropic residual metric

Let $A = (a_i)_{i=1}^n$ and $B = (b_j)_{j=1}^n$ be labelled planar point lists; repeated points are permitted. Fix one symmetric positive-definite matrix $W \in \mathbb{R}^{2 \times 2}$, expressed in target/world coordinates and shared

by every residual. With $R_\theta = \cos \theta I + \sin \theta J$, we consider

$$\min_{\substack{\theta \in \mathbb{R}/(2\pi\mathbb{Z}) \\ t \in \mathbb{R}^2, \pi \in S_n}} \sum_{i=1}^n (R_\theta a_i + t - b_{\pi(i)})^T W (R_\theta a_i + t - b_{\pi(i)}), \quad (\text{B.1})$$

where $J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$. Thus the source rotates, the target and metric remain fixed, and the allowed transformation is still a Euclidean rigid motion.

Theorem B.1. *For $n \geq 2$, at most $n(n-1)$ permutations are uniquely optimal at some angle in (B.1). This bound is sharp for every fixed positive-definite W . The correlation polygon defined below has $V \leq n(n-1)$ distinct vertices, including degenerate inputs. For rational A, B, W , a global optimizer has an exact algebraic representation of degree at most four and is computable in deterministic polynomial bit time. Its combinatorial phase uses at most $2V$ exact assignment queries and $O(n^5)$ rational arithmetic operations; the remaining phase uses $O(V)$ polynomial solves and comparisons of bounded algebraic degree. After eliminating translation and permutation, the angular cost is either constant or has only strict local minima, at most $2V$ modulo a full turn.*

Centering and matching count. Put $x_i = a_i - \bar{a}$ and $y_j = b_j - \bar{b}$. Differentiation with respect to translation, using invertibility of W and the full bijection, gives the unique optimum

$$t = \bar{b} - R_\theta \bar{a}. \quad (\text{B.2})$$

For each permutation define

$$\begin{aligned} p_\pi &= \sum_i x_i^T W y_{\pi(i)}, & q_\pi &= \sum_i (Jx_i)^T W y_{\pi(i)}, \\ Q &= \text{conv}\{(p_\pi, q_\pi) : \pi \in S_n\}. \end{aligned} \quad (\text{B.3})$$

Write

$$\begin{aligned} A_0 &= \sum_i x_i^T W x_i, & B_0 &= \sum_i (Jx_i)^T W (Jx_i), \\ C_0 &= \sum_i x_i^T W Jx_i, & T_0 &= \sum_j y_j^T W y_j, \end{aligned}$$

and set $g_0 = (A_0 + B_0)/2 + T_0$, $\alpha = (A_0 - B_0)/2$, $\beta = C_0$. Expansion of the centered cost gives

$$F_\pi(\theta) = g_0 + \alpha \cos 2\theta + \beta \sin 2\theta - 2p_\pi \cos \theta - 2q_\pi \sin \theta. \quad (\text{B.4})$$

The second harmonic is independent of the permutation. Under the usual identification with $\mathbb{C}$, putting $y'_j = Wy_j$ gives $p_\pi + iq_\pi = \sum_i \bar{x}_i y'_{\pi(i)}$. Consequently all matching comparisons coincide with the rotational correlation problem for (x_i) and (y'_j) . Theorems 3.6 and A.2 supply the matching and distinct-vertex bounds. For sharpness, transport any equality example (A, B') through $B = W^{-1}B'$; this preserves every comparison, and preserves rationality when W and the example are rational. This argument concerns matching count: the second harmonic generally prevents the Euclidean farthest-vertex rule from solving the whole optimization. Whitening also changes the allowed rotation group.

An exact finite candidate set. Retain one realizing permutation for every vertex $v = (p, q)$ of Q , and let F_v denote (B.4) with those coefficients. Support maximization yields

$$f(\theta) = \min_{v \in \text{vert } Q} F_v(\theta), \quad \min_\theta f(\theta) = \min_v \min_\theta F_v(\theta). \quad (\text{B.5})$$

Thus each branch may be minimized on the entire circle, without testing its stationary points against a normal cone. A globally minimizing pair (v, θ) also supplies a valid optimal permutation: a lower competing branch at that angle would contradict minimality over all pairs.

Use $z = \tan(\theta/2)$, with $(c, s) = ((1 - z^2)/(1 + z^2), 2z/(1 + z^2))$. Direct differentiation gives

$$\begin{aligned} \frac{(1 + z^2)^2}{2} F'_v(\theta) &= P_v(z), \\ P_v(z) &= (\beta + q)z^4 + (4\alpha + 2p)z^3 - 6\beta z^2 \\ &\quad + (2p - 4\alpha)z + (\beta - q). \end{aligned} \tag{B.6}$$

For nonzero P_v , isolate all its distinct real roots, including roots with even or higher multiplicity, and evaluate their costs together with the omitted angle $(c, s) = (-1, 0)$. These at most five candidates contain a branch minimum. If P_v vanishes identically, coefficient comparison gives $\alpha = \beta = p = q = 0$; the branch is constant and the identity rotation suffices. Translation follows from (B.2).

Elimination, ties, and bit complexity. In the half-angle coordinate the cost is $N_v(z)/(1 + z^2)^2$, where

$$\begin{aligned} N_v(z) &= g_0(1 + z^2)^2 + \alpha(1 - 6z^2 + z^4) + 4\beta z(1 - z^2) \\ &\quad - 2p(1 - z^4) - 4qz(1 + z^2). \end{aligned} \tag{B.7}$$

One must remove denominator factors before eliminating z . Indeed, in an isotropic branch $P_v = (1 + z^2)(qz^2 + 2pz - q)$, and the raw resultant with $N_v - u(1 + z^2)^2$ can vanish identically because of the complex roots $z = \pm i$. Repeatedly divide P_v by $\gcd(P_v, 1 + z^2)$, then take its monic square-free part P_v^* . This preserves precisely the distinct real stationary roots and makes P_v^* coprime to the denominator. Consequently

$$H_v(u) = \text{Res}_z(P_v^*(z), N_v(z) - u(1 + z^2)^2) \tag{B.8}$$

is nonzero and has degree at most four: its leading coefficient in u is a nonzero product of denominator values at roots of P_v^* . The identically zero derivative case is handled before this procedure.

Rational Sturm sequences isolate the real stationary and value roots; rational interval refinement identifies the value corresponding to each stationary root. Comparisons must include exact equality detection, for example polynomial gcds and root counts on intersecting isolating intervals. Refinement alone cannot terminate on equal values. It suffices to retain one minimizing representative; neither this procedure nor support enumeration lists every tied permutation.

If L is the input encoding length, centering and multiplication by W produce rational data of polynomial bit length. The support procedure of Theorem A.2, applied to x_i and Wy_j , therefore retains its $2V$ query and $O(n^5)$ arithmetic bounds, including a point or segment. The quartics and resultants have bounded degree and polynomial coefficient length. Standard root separation and exact univariate arithmetic give polynomial bit complexity for the additional $O(V)$ solves and comparisons; these comparisons are not ordinary unit-cost rational operations. At a selected stationary root, c, s, t and the objective belong to $\mathbb{Q}(z)$, of degree at most four. The omitted-angle candidate is rational. The output is an algebraic rotation matrix, not a transcendental angle. This completes the algorithmic assertion.

Local minima. Writing $h_Q(u) = \max_{v \in Q} v \cdot u$, the envelope is $f(\theta) = g_0 + \alpha \cos 2\theta + \beta \sin 2\theta - 2h_Q(\cos \theta, \sin \theta)$. Between support switches, f agrees with one analytic vertex branch. A nonconstant trigonometric polynomial of degree at most two has at most four derivative zeros, counting multiplicity, on the circle. Its strict minima alternate with maxima, so each branch contributes at

most two. Counting on the whole circle avoids any issue with an active interval crossing angle zero. At a genuine support switch, $h'_{Q,+} - h'_{Q,-} > 0$, including both switches of a nontrivial segment. The cost therefore has $f'_+ < f'_-$, incompatible with the necessary minimum condition $f'_- \leq 0 \leq f'_+$. Switches contribute no minima.

The average of all permutation correlation vectors is zero, since both point lists are centered. This positive average puts 0 in relint Q . A nontrivial Q consequently has no zero vertex, hence no constant vertex branch by the coefficient condition above. Every local minimum therefore lies on a nonconstant analytic branch and is strict. If Q is a singleton, it equals $\{0\}$; the cost is constant or a nonconstant second harmonic with exactly two strict minima. This proves the stated dichotomy and bound. A constant cost has no strict or isolated minima, although every angle is a non-strict minimizer.

Example B.2. Take $A = \{(1, 0), (-1, 0)\}$, $B = \{(0, 1/4), (0, -1/4)\}$, and $W = \text{diag}(1, 2)$. Then Q has vertices $(0, \pm 1)$, but

$$f(\theta) = \frac{9}{4} + 2 \sin^2 \theta - 2|\sin \theta| = \frac{7}{4} + 2\left(|\sin \theta| - \frac{1}{2}\right)^2.$$

There are four strict global minima, at $\pi/6, 5\pi/6, 7\pi/6, 11\pi/6$. Thus the Euclidean bound V fails, and $2V$ is attained here. This does not establish sharpness for every larger n . The Euclidean radial choice $\theta = \pi/2$ instead costs $9/4$, exceeding the optimum $7/4$.

Boundary cases and scope. For $n = 1$, translation fits the sole point exactly: every angle is optimal, and there is one permutation. Reflections require a second run with source SA , $S = \text{diag}(1, -1)$, because every orientation-reversing orthogonal map is $R_\theta S$; comparing the two optima preserves the asymptotic complexity. For $W = \lambda I$, the quadratic Euclidean representation of Theorem A.2 remains available. The extension assumes one known, fixed, common metric and full bijection. It asserts no corresponding theorem for point-dependent or pair-dependent metrics, rotation-dependent covariance, unknown metrics, partial matching, outliers, additional scalar weights, scaling, affine maps, or rotations in higher dimensions.

C. Why arbitrary masses are different

The permutation model is essential. Consider the transportation polytope with positive row masses r and column masses c , both summing to m . For a rotation, minimizing squared transport cost is equivalent to maximizing

$$\sum_{i,j} X_{ij} (d_0(a_i \cdot b_j) + d_1(a_{ix}b_{jy} - a_{iy}b_{jx})), \quad X\mathbf{1} = r, \quad X^\top \mathbf{1} = c, \quad X \geq 0,$$

where the nonzero direction d can be normalized to unit length. Dividing all masses and plans by m would give the probability-mass convention without changing which plans are optimal.

Proposition C.1. *There are three sources and three targets, with fixed positive integer marginals, for which rotation exposes at least nine distinct unique optimal transport plans. In particular, the six-permutation bound for three equal-mass matched pairs does not extend unchanged to this model.*

Proof. Take

$$A = \begin{pmatrix} -2 & 7 \\ -5 & -1 \\ -9 & -7 \end{pmatrix}, \quad B = \begin{pmatrix} -3 & 8 \\ -6 & -5 \\ -1 & -6 \end{pmatrix}, \quad r = (2, 5, 11), \quad c = (3, 6, 9).$$

The nine directions and plans in Table 2 have the required marginals. Each support is a connected five-edge tree in the bipartite graph of three rows and three columns. Set the first row dual potential to zero, then propagate the equations $u_i + v_j = C_{ij}$ along this tree, where C is the displayed directional score matrix. The resulting potentials satisfy $u_i + v_j > C_{ij}$ at all four off-support entries; the final column gives the minimum of these four integer slacks. Thus every optimal plan is supported on this tree. The marginals determine its entries uniquely by successively removing leaves of the tree. All nine listed plans are distinct, proving the assertion. The supplied exact certificates contain every dual potential and every slack, and the checker verifies these equalities and inequalities without enumerating the transportation polytope. $\square$

Table 2: Nine strictly exposed transport plans. Each triple separated by a semicolon is one row of X . Slacks refer to the unnormalized integer direction.

d_0	d_1	Rows of X	Minimum slack
-111	69	(0, 2, 0); (0, 4, 1); (3, 0, 8)	3744
-120	-28	(0, 1, 1); (0, 5, 0); (3, 0, 8)	364
-247	-91	(0, 0, 2); (0, 5, 0); (3, 1, 7)	364
82	-434	(0, 0, 2); (0, 0, 5); (3, 6, 2)	15904
320	-440	(2, 0, 0); (0, 0, 5); (1, 6, 4)	2800
204	-20	(2, 0, 0); (1, 0, 4); (0, 6, 5)	3536
106	282	(2, 0, 0); (1, 4, 0); (0, 2, 9)	8104
-98	406	(0, 2, 0); (3, 2, 0); (0, 2, 9)	4984
-136	256	(0, 2, 0); (1, 4, 0); (2, 0, 9)	3536

This example does not rule out a larger polynomial bound for transportation. Nor does replacing integer masses by repeated unit points transfer the original complexity statement: the expanded number of points depends on the numerical masses, not merely their binary encoding length. The common metric of Appendix B changes the geometry of every residual in the same way; it is not a way to incorporate arbitrary point masses or pair-dependent quadratic forms.

D. Structured bounds for unequal-mass transport

The counterexample in Appendix C leaves a separate complexity question. This appendix gives a residual-flow reduction, special-case polynomial bounds, and all-size hull-terminal sharpness. None is a polynomial bound for arbitrary planar transport with both support sizes growing and arbitrary positive marginals.

D.1. A support-only shadow bound

Let $A = (a_1, \dots, a_m)$ and $B = (b_1, \dots, b_n)$ have positive marginals α, β of equal total mass. Define

$$T(\alpha, \beta) = \{X \geq 0 : X\mathbf{1} = \alpha, X^\top \mathbf{1} = \beta\}, \quad P = \left\{ \sum_{ij} X_{ij} \bar{a}_i b_j : X \in T(\alpha, \beta) \right\}.$$

If U plans are uniquely optimal at some rotation and V is the number of vertices of P , then $U \leq V$: a unique plan must map to a strictly exposed vertex with a singleton fiber. The converse can fail.

Proposition D.1. For $m, n \geq 2$,

$$V \leq \sum_{k=2}^{\min(m,n)} \binom{m}{k} \binom{n}{k} k(k-1) = \frac{(m+n-2)!}{(m-2)!(n-2)!}. \quad (\text{D.1})$$

If one support size is one, the plan is fixed and $U = V = 1$. The bound is independent of the magnitudes and denominators of the masses.

Proof. An edge of a transportation polytope is parallel to an alternating simple cycle of its bipartite support graph. At a noncollapsed edge of its planar shadow, the exposed preimage face contains an original edge with nonzero projection: otherwise its connected edge graph would have constant image. Suppose its cycle uses row set I and column set J , both of size k . An optimal transportation dual is tight on this cycle. Restricted to I, J , it certifies that the two alternating perfect matchings are optimal in the corresponding unweighted assignment problem. Their projected correlations are distinct. Thus the shadow-edge normal is also an edge-normal direction of that subset assignment shadow.

By Lemma A.1, a k -by- k assignment shadow has at most $k(k - 1)$ such directions, including the two directions for a segment. Taking the union over I, J proves the sum bound for a two-dimensional P ; a segment or point satisfies it as well. Finally, $k(k - 1) \binom{m}{k} = m(m - 1) \binom{m-2}{k-2}$ and Vandermonde's identity give $m(m - 1) \binom{m+n-2}{m}$, equal to the displayed factorial ratio. $\square$

For fixed m this is $O_m(n^m)$, but for balanced growing supports it is exponential. If the source points are distinct and collinear, a sharper $U \leq n(n - 1)$ holds independently of m and the masses: one-dimensional rearrangement gives the unique monotone quantile coupling on each strict projection-order cell, and each target pair ties at two antipodal angles. A unique plan at an exceptional angle persists nearby and is already counted. Interchanging the supports gives the analogous bound. With both supports collinear there are at most two unique plans.

D.2. One unit background with one mass defect

The following reduction uses arbitrary costs, not geometric assumptions.

Proposition D.2. *Suppose the identity is the unique minimum-cost unweighted assignment for an N -by- N matrix C . Fix $s \neq t$ and $0 < \varepsilon < 1$, and take*

$$\alpha = \mathbf{1} + \varepsilon e_s, \quad \beta = \mathbf{1} + \varepsilon e_t.$$

On the complete directed graph without loops put $w_{ij} = c_{ij} - c_{ii}$. The optimal transport plans are precisely the plans induced by convex combinations of shortest simple s -to- t paths. A plan is unique if and only if the shortest path is unique.

Proof. For $i \neq j$ put $f_{ij} = X_{ij}/\varepsilon$. The margins give

$$\text{out}_i(f) - \text{in}_i(f) = \mathbf{1}_{i=s} - \mathbf{1}_{i=t}, \quad X_{ii} = 1 + \varepsilon \mathbf{1}_{i=s} - \varepsilon \text{out}_i(f),$$

and the objective is

$$\sum_i c_{ii} + \varepsilon c_{ss} + \varepsilon \sum_{i \neq j} f_{ij} w_{ij}. \quad (\text{D.2})$$

Every nontrivial simple directed cycle has positive w -cost, since replacing the identity by that cycle is a worse permutation. Decompose the nonnegative unit flow into paths and cycles. Removing cycles increases the displayed diagonal entries, preserving feasibility, and strictly improves the cost. A cycle-free unit flow is a convex combination of simple paths. Conversely every such combination has outflow at most one at every vertex, hence gives nonnegative diagonals and a feasible plan. Equation (D.2) proves optimality. A simple path places mass ε on its arcs, mass $1 - \varepsilon$ at the diagonals of its internal vertices, and mass 1 on all other diagonals, including s, t . Distinct paths give distinct plans, proving the uniqueness assertion. $\square$

In particular $\varepsilon = 1/2$ suffices throughout. Increasing denominator size or mass ratio is not needed to obtain the residual-path mechanism.

D.3. Self-registration and a fixed Delaunay graph

Take $A = B = (p_1, \dots, p_N)$, with distinct planar sites, and rotate targets clockwise by θ . Write $\det(p, q) = p_x q_y - p_y q_x$. Direct expansion gives

$$w_{ij}(\theta) = \cos \theta |p_j - p_i|^2 - 2 \sin \theta \det(p_i, p_j) + (1 - \cos \theta)(|p_j|^2 - |p_i|^2).$$

The last term telescopes along paths. Set $\lambda = \tan \theta$ and

$$c_\lambda(i, j) = |p_j - p_i|^2 - 2\lambda \det(p_i, p_j), \quad E(P) = \sum_{ij \in P} |p_j - p_i|^2, \quad A(P) = \sum_{ij \in P} \det(p_i, p_j). \quad (\text{D.3})$$

Thus path comparison reduces to $E(P) - 2\lambda A(P)$ while $\cos \theta > 0$. Here A is twice the signed area after adding a fixed closing curve, not the area itself.

Lemma D.3. *The identity assignment is uniquely optimal for $|\theta| < \pi/N$. Every nontrivial directed cycle has positive c_λ -cost on*

$$I_N = (-\tan(\pi/N), \tan(\pi/N)), \quad I_2 = \mathbb{R}.$$

Proof. For a k -cycle of complex sites $z_0, \dots, z_{k-1}$, with discrete Fourier coefficients $\widehat{z}_r = k^{-1} \sum_j z_j \exp(-2\pi i r j/k)$,

$$E = k \sum_r 4 \sin^2(\pi r/k) |\widehat{z}_r|^2, \quad A = k \sum_r \sin(2\pi r/k) |\widehat{z}_r|^2.$$

It follows that $|A| \leq \frac{1}{2} \cot(\pi/k) E$. Since $E > 0$, $E - 2\lambda A > 0$ on I_N for $k \leq N$; the two-cycle has $A = 0$. The telescoping term vanishes on a cycle and $\cos \theta > 0$ in the stated angular range. Decompose any nonidentity permutation into cycles. $\square$

Proposition D.4 (Strict empty-disk condition). *For $\lambda \in I_N$, every edge $i \rightarrow j$ of a unique shortest path has no third site in the closed disk of center and squared radius*

$$o_{ij}(\lambda) = \frac{p_i + p_j}{2} - \frac{\lambda}{2} J(p_j - p_i), \quad r_{ij}(\lambda)^2 = \frac{1 + \lambda^2}{4} |p_j - p_i|^2. \quad (\text{D.4})$$

All such edges therefore lie in a fixed planar strict Delaunay graph, independently of λ . Their necessary eligibility set is the open interval, possibly empty or unbounded,

$$I_N \cap \bigcap_{k \notin \{i, j\}} \{\lambda : (p_k - p_i) \cdot (p_k - p_j) + \lambda \det(p_j - p_i, p_k - p_i) > 0\}. \quad (\text{D.5})$$

Proof. Inserting k between i, j changes the cost by

$$c_\lambda(i, k) + c_\lambda(k, j) - c_\lambda(i, j) = 2(|p_k - o_{ij}(\lambda)|^2 - r_{ij}(\lambda)^2).$$

If this is negative, insertion gives a cheaper walk; removing its positive cycles gives a cheaper simple path. If it is zero and k was absent from the path, insertion gives a distinct equal-cost simple path. If k was already present, removing the resulting nontrivial positive cycle strictly improves the walk. All three possibilities contradict unique optimality. Expansion gives (D.5), an intersection of strict linear inequalities in one variable.

For completeness, lift sites to $(p_i, |p_i|^2)$. A circle through exactly two sites with all others outside its closed disk defines a lower supporting plane whose contact face is their lifted edge. Projections of these lower-hull edges do not cross. Triangulating the polygonal lower faces gives one fixed Delaunay triangulation containing every such edge. For collinear data the strict edges are consecutive-site edges. This is the classical empty-circle construction, not a new triangulation theorem. $\square$

The condition is necessary, not sufficient, for an edge to participate in a globally shortest path. For noncollinear data the containing triangulation has $3N - h - 3$ edges and $F = 2N - h - 2$ bounded faces, where h counts all sites on the convex-hull boundary, including any collinear boundary sites.

D.4. A quadratic bound from integer face potentials

Theorem D.5. *For distinct noncollinear self-registration sites and arbitrary distinct terminals s, t , at most $4F^2 + 1$ simple s -to- t paths are uniquely optimal on I_N , where $F = 2N - h - 2$. The same bound holds for unique one-defect transport plans on $|\theta| < \pi/N$.*

Proof. Fix a triangulation containing all strict Delaunay edges, orient its undirected edges arbitrarily, and choose a simple reference s -to- t flow f^0 . Bounded-face boundaries form an integral basis for integer circulations in a connected plane graph. Hence every integral net flow with divergence $e_s - e_t$ has the unique form

$$f_e = f_e^0 + \psi_{L(e)} - \psi_{R(e)}, \quad \psi \in \mathbb{Z}^F, \quad \psi_{\text{outer}} = 0,$$

where $L(e), R(e)$ are the faces to the left and right of an oriented edge. For example, integer face potentials are obtained by integrating the circulation along the dual graph; path independence is exactly its zero-divergence condition.

Put $\ell_e = |p_j - p_i|^2$ for an edge oriented $i \rightarrow j$. Cancelling opposite arcs always reduces cost, since their total cost is $2\ell_e > 0$. The integral net-flow objective, up to a common term, is therefore

$$H_\lambda(\psi) = K(\psi) - 4\lambda \sum_f a_f \psi_f, \quad K(\psi) = \sum_e \ell_e |f_e^0 + \psi_{L(e)} - \psi_{R(e)}|, \quad (\text{D.6})$$

where $a_f > 0$ is the Euclidean area of bounded face f . Indeed, the cross products around its counterclockwise boundary sum to $2a_f$; the omitted term is $-2\lambda A(f^0)$.

Every integral unit flow decomposes into one simple s -to- t path and integer directed cycles. By Lemma D.3, deleting cycles strictly reduces cost. Thus the minimum in (D.6) exists among finitely many simple paths, and every minimizer is a path. No coercivity assumption on the infinite lattice is needed.

The function K is lattice-submodular. Each summand has the form $q(c + x - y)$ with q convex. Its submodular inequality is equality when the two coordinate pairs have the same order; in the opposite-order case it follows by moving two arguments with fixed sum closer together. Thus, with componentwise minimum and maximum denoted by $\wedge, \vee$,

$$K(\psi) + K(\phi) \geq K(\psi \wedge \phi) + K(\psi \vee \phi).$$

If ψ minimizes at λ_1 and ϕ at $\lambda_2 > \lambda_1$, compare them with $\psi \wedge \phi$ and $\psi \vee \phi$, respectively. Their optimality inequalities give

$$0 \leq K(\psi) + K(\phi) - K(\psi \wedge \phi) - K(\psi \vee \phi) \leq -4(\lambda_2 - \lambda_1) \sum_f a_f (\psi_f - \phi_f)_+.$$

It follows that $\psi \leq \phi$ coordinatewise.

For a simple path, $|f_e - f_e^0| \leq 2$. Every bounded face is connected to the outer face by a dual path of at most F edges, so $|\psi_f| \leq 2F$. Consequently $\sum_f \psi_f$ is an integer in $[-2F^2, 2F^2]$ and increases by at least one at every change to a distinct unique path. There are at most $4F^2 + 1$ such paths in the triangulation. Every unique complete-graph optimum belongs to the triangulation by Proposition D.4, so the bound holds there too. Apply Proposition D.2 for transport. $\square$

The argument applies the classical monotone-lattice minimization principle of Topkis [22], with the needed inequality supplied explicitly. It does not assume that either terminal is incident to the unbounded face, or change the geometric outer face.

Theorem D.6 (Binary perturbations of balanced masses). *For the same distinct noncollinear self-registration sites, let $u, v \in \{0, 1\}^N$ satisfy $\sum_i u_i = \sum_i v_i$, and put*

$$K = \sum_i (u_i - v_i)_+, \quad \alpha = \mathbf{1} + \varepsilon u, \quad \beta = \mathbf{1} + \varepsilon v, \quad 0 < \varepsilon < \frac{1}{\max(1, K)}.$$

On $|\theta| < \pi/N$, at most $4KF^2 + 1$ transport plans are uniquely optimal. If $K = 0$, the unique plan is the fixed diagonal plan. Since $K \leq N/2$, this is an $O(N^3)$ bound, independent of the denominator of ε .

Proof. For $i \neq j$ set $f_{ij} = X_{ij}/\varepsilon$. The constraints become

$$\operatorname{div} f = u - v, \quad X_{ii} = 1 + \varepsilon u_i - \varepsilon \operatorname{out}_i(f).$$

As in (D.2), the transport objective differs from $\varepsilon \sum_{i \neq j} f_{ij}(c_{ij} - c_{ii})$ by a constant. After removing vertex-potential terms, which are constant at fixed divergence, comparison reduces to $\sum f_{ij} c_\lambda(i, j)$.

Every nonnegative flow of this divergence decomposes into source-to-sink paths of total mass K and directed cycles. Removing cycles strictly decreases cost and increases the corresponding diagonal entries. Every cycle-free flow has $\operatorname{out}_i(f) \leq K$, so it gives a feasible transport plan since $1 - \varepsilon K > 0$. Thus the transport optimum is exactly the unconstrained minimum-cost flow optimum, and all optimal flows are cycle-free. Existence follows without an unbounded-polyhedron assumption: cycle removal puts any candidate into the compact box $0 \leq f_{ij} \leq K$.

We give the needed integrality argument explicitly. Choose an optimal flow with minimum support size. If its undirected support has a cycle, perturb along the corresponding signed circulation in both directions. For sufficiently small perturbations both flows remain nonnegative; optimality forces zero cost change. Pushing to the first vanishing entry then reduces the support, a contradiction. Its support is therefore a forest, and integer balances imply integer edge flows by successive leaf elimination. In particular an integral optimum exists, and every unique optimum is integral by the same support-cycle argument.

The strict empty-disk condition also applies to every positive edge of a unique flow. Divert a small amount from $i \rightarrow j$ through a third site k . A negative insertion cost, followed by cycle removal, strictly improves the flow. A zero insertion cost either creates a positive-cost cycle, whose removal again improves it, or yields a distinct cycle-free equal-cost feasible flow. Both contradict uniqueness. Thus every unique optimal flow is supported on the fixed strict Delaunay graph.

On a containing triangulation choose an integral reference flow f^0 of divergence $u - v$ and edge magnitudes at most K , by routing the K source units along simple paths. The integer face-potential representation and energy (D.6) remain valid with this reference flow. Its offset does not change submodularity. Therefore integer optima at increasing parameters have coordinatewise nondecreasing potentials. Every optimum is cycle-free and has edge magnitudes at most K ; hence $|f_e - f_e^0| \leq 2K$ and $|\psi_f| \leq 2KF$. The sum of the F integer potentials lies in $[-2KF^2, 2KF^2]$, giving at most $4KF^2 + 1$ distinct unique optima. All unique complete-graph optima are among these optima. For $K = 0$ every off-diagonal flow is a circulation, so its positive cycles are removed and only the diagonal plan remains. $\square$

The small-perturbation restriction is essential to this reduction: without it, the diagonal nonnegativity constraints may impose capacities on the residual flows. The result does not cover arbitrary positive marginals. We next allow real amplitudes, then sharpen the one-defect hull-terminal count.

D.5. A fixed number of arbitrary real perturbation amplitudes

The integer-mask argument extends immediately from binary masks to nonnegative integer u, v : the bound is still $4KF^2 + 1$, where $K = \sum_i (u_i - v_i)_+$. If all entries are at most the integer M , then $K \leq NM/2$ and the count is $O(MN^3)$. Clearing denominators does not turn this into a denominator-independent theorem for arbitrary real perturbations. The following different counting step does so when the number of amplitudes and the integer coefficients are bounded.

Theorem D.7 (Fixed real-amplitude complexity). *For the distinct noncollinear self-registration sites of Theorem D.5, let $u, v \in \mathbb{R}_{\geq 0}^N$ have equal sums. Put*

$$d = u - v, \quad K = \sum_i (d_i)_+, \quad \alpha = \mathbf{1} + \varepsilon u, \quad \beta = \mathbf{1} + \varepsilon v, \quad 0 < \varepsilon < \frac{1}{\max(1, K)}.$$

Suppose that, for positive integers q, M ,

$$d_i = \sum_{r=1}^q c_{ir} \eta_r, \quad c_{ir} \in \mathbb{Z}, \quad |c_{ir}| \leq M, \quad \eta_r \in \mathbb{R}.$$

No rational independence of the amplitudes is assumed, and ε need not be rational. On $|\theta| < \pi/N$, the number of distinct uniquely optimal transport plans is at most

$$1 + F[(4NMF + 1)^q - 1], \quad F = 2N - h - 2. \quad (\text{D.7})$$

For fixed q, M this is polynomial in N , independently of the numerical amplitudes and all denominators. If $K = 0$, the unique plan is diagonal.

Proof. The cycle-removal and feasibility argument in Theorem D.6 uses only $u, v \geq 0$ and $\varepsilon K < 1$, not integrality. Thus the transport optimum equals unconstrained transshipment of divergence d . A minimum exists after cycle removal into the compact box of arc flows between zero and K . Every unique optimal real flow has undirected forest support: a supported cycle would permit signed circulation perturbations of both signs; optimality forces their cost change to vanish, violating uniqueness. Each forest-edge flow is a signed sum of divergences over one side of the corresponding cut in its tree component.

The strict insertion argument also does not require integrality. Consequently every unique optimum belongs to a fixed triangulation containing the strict Delaunay graph. Fix a spanning tree of that triangulation and its unique real reference flow f^0 of divergence d . Normalized face potentials now range over $\mathbb{R}^F$, with outer value zero. The energy is still (D.6). Its submodularity holds for real coordinates, so minimizers ψ, ϕ at $\lambda_1 < \lambda_2$ satisfy

$$0 \leq G(\psi) + G(\phi) - G(\psi \wedge \phi) - G(\psi \vee \phi) \leq -4(\lambda_2 - \lambda_1) \sum_f a_f (\psi_f - \phi_f)_+,$$

where G is the absolute-value part of that energy and $a_f > 0$. Thus $\psi \leq \phi$ coordinatewise; the real-valued monotonicity step requires no discretization.

Each edge value of f^0 , and each edge value of a unique forest optimum f , is a signed subset sum of the d_i . It therefore admits a representation in the amplitudes η_r with integer coefficients of absolute value at most NM . For $f - f^0$ the bound is $2NM$. Fix a dual path of at most F edges from the outer face to each bounded face. Integrating $f - f^0$ along it expresses the actual real potential as

$$\psi_f = \sum_{r=1}^q z_{fr} \eta_r, \quad z_{fr} \in \mathbb{Z}, \quad |z_{fr}| \leq 2NMF.$$

The representations need not be unique. Nor must the coefficient fields separately be circulations: integration along the fixed path computes the actual real potential. Rational dependence among the amplitudes therefore does not affect the bound.

Each coordinate of a unique optimum lies in a fixed set of at most $L = (4NMF + 1)^q$ real numbers. Order one parameter witness for each distinct unique plan. Their potential vectors form a nondecreasing chain, and distinct plans give distinct vectors. Each new vector strictly increases at least one coordinate; each of the F coordinates can increase at most $L - 1$ times. This gives (D.7). If $K = 0$, all residual flows are circulations and positive cycle costs force zero flow. $\square$

For example, if each entry of u and v is chosen from zero and q arbitrary real amplitudes, then $M = 1$. With fixed $q = 2$ the bound is $O(N^5)$, even for irrational or arbitrarily close amplitudes. The common component shared by u and v is unrestricted; only their difference needs the displayed representation. If q grows with N , the estimate is not uniformly polynomial. This is not a solution for arbitrary real marginal vectors or for capacity-binding residual flows. As in Remark D.10, the proof holds on any open interval where every directed geometric cycle has strictly positive cost.

D.6. Hull terminals give a linear phase bound

Theorem D.8. *Suppose the sites are noncollinear and both p_s, p_t lie on the convex-hull boundary. On I_N , at most $2N - h - 1$ simple s -to- t paths are uniquely optimal for (D.3). Hence the same bound holds for unique one-defect self-registration transport plans on $|\theta| < \pi/N$. If all sites are in convex position the bound is $N - 1$.*

Proof. Fix a triangulation containing the strict Delaunay graph. Every unique optimum in the complete graph remains unique in this subgraph. Add a simple exterior arc $\gamma : t \rightarrow s$, disjoint from the triangulation except at its endpoints, with a bounded face f_+ to its left and the unbounded face f_- to its right. The F original bounded faces are the optional faces. Consider face sets S containing f_+ but not f_- . Let $f(S)$ be the sum of squared lengths of original edges separating S from its complement, and $a(S)$ its area. A simple path together with γ bounds one such face region. Green's formula gives

$$\text{cost}_\lambda(P) = f(S) - 4\lambda a(S) + 2\lambda A(\gamma), \quad A(\gamma) = \int_\gamma (x dy - y dx). \quad (\text{D.8})$$

Minimizing $f(S) - 4\lambda a(S)$ over all these face sets is an exact relaxation of the path problem. Indeed, orient the cut boundary with the included faces on its left. It is a directed Eulerian subgraph containing γ , even if the face region is disconnected or has pinched boundary vertices. Decompose it into simple directed cycles; the cycle containing γ yields a simple s -to- t path after that arc is removed. Every remaining cycle has strictly positive cost by Lemma D.3. Boundary costs add, so such a cut cannot improve on the path alone, and it is strictly worse if other cycles remain. Conversely each simple path supplies its bounded face region and equality in (D.8). This also handles holes and islands.

The positive weighted cut function f is submodular. Let S, T be optimal face regions at $\lambda_1 < \lambda_2$. Compare S with $S \cap T$ at λ_1 , and T with $S \cup T$ at λ_2 . Adding their optimality inequalities and using area additivity gives

$$0 \leq f(S) + f(T) - f(S \cap T) - f(S \cup T) \leq -4(\lambda_2 - \lambda_1)a(S \setminus T).$$

Every face has positive area, so $S \subseteq T$. The mandatory faces agree in all admissible sets. Each change to a different unique path therefore adds at least one of the F optional faces. At most $F + 1 = 2N - h - 1$ distinct unique paths occur. Apply Proposition D.2 for transport. $\square$

For collinear sites the consecutive-site path between the endpoints is uniquely shortest, independently of λ , so its separate bound is one. Both theorems concern the stated angular interval and mass pattern. Interior terminals have the quadratic bound, not the linear hull bound. Neither theorem counts arbitrary marginals or all tied paths. All-size sharpness below concerns general positive-cycle intervals, not the universal interval I_N .

Example D.9. The linear bound is attained for four convex-position sites. Take $s = (-2, 0)$, $t = (2, 0)$, $u = (0, 3)$ and $d = (0, -3)$. The five simple s -to- t paths have energy–area pairs

$$(E, A) = (16, 0), \quad (26, -12), \quad (26, 12), \quad (62, 0), \quad (62, 0),$$

respectively for the direct path, the routes through u and through d , and the two routes through both. The last two are always dominated. The first three are uniquely optimal on successive subintervals separated by $\lambda = -5/12$ and $\lambda = 5/12$, both in $I_4 = (-1, 1)$. Thus three unique paths occur, equal to $N - 1$. Via [Proposition D.2](#), the same example uses only masses 1 and $3/2$. This is a four-point equality example, not an asymptotic sharpness claim.

Remark D.10. The disk, face-potential, and hull-region arguments need only strict positivity of every directed cycle cost $E(C) - 2\lambda A(C)$. They therefore hold on any open parameter interval satisfying that condition, not just the guaranteed interval I_N . The condition is an intersection of finitely many strict linear inequalities in λ ; no algorithm for enumerating those cycles is required or asserted here.

D.7. All-size sharpness on a certified positive-cycle interval

The extension in [Remark D.10](#) is sharp for every support size. The interval in the following construction is verified directly; it is larger than, and must not be confused with, the universal small-angle interval I_N .

Theorem D.11 (Sharp parabolic fan). *For every $N \geq 3$, there are N integer sites in strictly convex position and two fixed hull terminals for which exactly $N - 1$ simple paths are uniquely optimal at some parameter of one positive-cycle interval for $E(P) - 2\lambda A(P)$. They give $N - 1$ distinct unique self-registration transport plans with masses in $\{1, 3/2\}$ on a certified interval where the unweighted identity assignment is uniquely optimal. For $N = 2$ the corresponding sharp count is one.*

Proof. For $N \geq 3$, set

$$M = 32N^2, \quad x_0 = 0, \quad x_i = M + i \quad (1 \leq i < N), \quad p_i = (x_i, x_i^2), \quad L = 3(M + N). \quad (\text{D.9})$$

Take $s = 0$, $t = N - 1$, and $-L < \lambda < L$. Each site is exposed by its parabola tangent: at another site the separation from $y = 2x_i x - x_i^2$ is $(x_j - x_i)^2 > 0$. The convex-hull order is therefore $0, 1, \dots, N - 1$, closed by the last-to-first edge.

We first certify every directed cycle on the whole closed interval $|\lambda| \leq L$. On an edge write $\delta = x_j - x_i$; then

$$|p_j - p_i|^2 = \delta^2 [1 + (x_i + x_j)^2].$$

Telescoping cubes around a directed cycle gives

$$A(C) = \sum x_i x_j (x_j - x_i) = -\frac{1}{3} \sum \delta^3. \quad (\text{D.10})$$

For a simple cycle avoiding 0, put $S = \sum \delta^2 > 0$. All its sites lie in the cluster, so $|\delta| \leq N$ and $x_i + x_j \geq 2M$. Hence

$$E(C) \geq 4M^2 S, \quad |A(C)| \leq NS/3,$$

and, using $L \leq 6M$,

$$E(C) - 2\lambda A(C) \geq 4M(M - N)S > 0.$$

For a simple cycle containing 0, its two incident edges contribute at least $2M^4$ to the energy and zero to the area sum. There are at most N other edges, each with cross product of absolute value at most $N(M + N)^2$. Consequently

$$\begin{aligned} E(C) - 2\lambda A(C) &\geq 2M^4 - 6N^2(M + N)^3 \\ &\geq 2M^3(M - 24N^2) > 0. \end{aligned}$$

All closed walks decompose into simple directed cycles, so this proves the required strict-positive-cycle condition.

The Delaunay structure is explicit. For $a = x_i$, $b = x_{i+1}$, the power of (x, x^2) relative to the circle through p_0, p_i, p_{i+1} is

$$x(x - a)(x - b)(x + a + b). \quad (\text{D.11})$$

This follows by expanding the circle equation restricted to $y = x^2$. At every other sampled site the expression is strictly positive, since there is no sampled coordinate between a and b . Thus the triangles $(0, i, i + 1)$ form a fan of strictly empty circumcircles. No three sites are collinear. Nor are four cocircular: restriction of any circle to the parabola is a monic quartic with zero cubic coefficient, whereas four distinct nonnegative roots would have positive sum.

Crucially, every complete-graph shortest path belongs to this fan, including at tie parameters. If a shortest-path edge had a third site inside its shifted disk, the insertion calculation in the proof of [Proposition D.4](#) would give a cheaper walk. Removing any resulting positive-cost cycles would give a cheaper simple path. Thus every shortest edge has a weakly empty disk; boundary sites are allowed in this argument.

Lift the sites to $(p_i, |p_i|^2)$. A weakly empty disk gives a lower supporting plane touching its two endpoint lifts. The strictly empty circles (D.11) give lower triangular faces, and their projections tile the convex hull. On each triangle its plane is the lower hull: it bounds every convex combination from below and the triangle's barycentric combinations attain the bound. These are therefore all the lower facets, with their edges and vertices. A lower supporting face containing two site lifts must belong to this fan. Hence every shortest-path edge is a fan edge, not merely every edge of a uniquely shortest path.

After a simple path leaves 0, it cannot revisit that hub. The remaining fan graph is the chain $1 - -2 - - \dots - -(N - 1)$, so all shortest paths lie in the complete candidate list

$$P_k = (0, k, k + 1, \dots, N - 1), \quad 1 \leq k < N.$$

For $1 \leq k \leq N - 2$, define

$$\tau_k = -\frac{1 + x_k^2 + x_k x_{k+1}}{x_{k+1}} = -\left(2x_k - 1 + \frac{2}{x_k + 1}\right). \quad (\text{D.12})$$

Direct subtraction yields

$$C_\lambda(P_k) - C_\lambda(P_{k+1}) = -2x_k x_{k+1}(\lambda - \tau_k), \quad C_\lambda = E - 2\lambda A.$$

The thresholds satisfy $\tau_{N-2} < \dots < \tau_1$, because the successive increments of $2x - 1 + 2/(x + 1)$ are $2 - 2/((x + 1)(x + 2)) > 0$. Moreover $|\tau_k| < 2(M + N) < L$. With $\tau_0 = L$ and $\tau_{N-1} = -L$, the path P_k is the unique global optimum on the nonempty interval (τ_k, τ_{k-1}) : successive candidate costs strictly decrease toward index k and strictly increase after it. At τ_k exactly P_k and P_{k+1} are optimal. This proves the count $N - 1$.

Finally let the target list rotate clockwise by $\theta = \arctan \lambda$, so $\cos \theta > 0$. Its residual squared-cost arcs satisfy

$$c_{ij}(\theta) - c_{ii}(\theta) = \cos \theta C_\lambda(i, j) + (1 - \cos \theta)(|p_j|^2 - |p_i|^2).$$

The last term telescopes around cycles and is fixed on paths with the specified endpoints. Every nonidentity assignment has positive residual cost by cycle decomposition, so identity is uniquely optimal throughout $|\theta| < \arctan L$. Apply [Proposition D.2](#) with $\epsilon = 1/2$, surplus source 0, and surplus target $N - 1$. The path sequence gives the asserted distinct unique plans. For $N = 2$ there is one simple path and all cycles have zero area and positive energy, giving the remaining case directly. $\square$

The construction certifies sharpness on a general identity-optimal positive-cycle interval, not all-size sharpness within the fixed universal interval $I_N = (-\tan(\pi/N), \tan(\pi/N))$.

D.8. A quasipolynomial bound for one-defect masses over all rotations

For comparison, the residual reduction also gives a whole-rotation bound without the self-registration or hull-terminal assumptions. We include an elementary derivation of the usual type of parametric shortest-path bound, not a novelty claim about general parametric graph algorithms.

Proposition D.12. *For two planar lists of size $N \geq 2$ and fixed masses $\alpha = \mathbf{1} + \epsilon e_s$, $\beta = \mathbf{1} + \epsilon e_t$, with $0 < \epsilon < 1$, the number of distinct transport plans uniquely optimal at some rotation is at most*

$$[N(N - 1) + 2](2N)^{\lceil \log_2(\max(1, N - 1)) \rceil} = N^{O(\log N)}. \quad (\text{D.13})$$

Proof. For affine edge costs on a complete directed N -vertex graph, add zero loops and let $D_k(i, j)$ be the minimum cost of walks with at most 2^k edges. It is concave piecewise affine in the parameter, and

$$D_k(i, j) = \min_u \{D_{k-1}(i, u) + D_{k-1}(u, j)\}.$$

If each function at level $k - 1$ has at most C_{k-1} affine pieces, each sum has at most $2C_{k-1}$ pieces. A concave piecewise-affine function is the minimum of the affine extensions of its pieces. Thus taking the minimum over N middle vertices gives $C_k \leq 2NC_{k-1}$, with $C_0 = 1$. On an interval where all nontrivial cycles have positive cost, a shortest walk is simple. Taking $k = \lceil \log_2(\max(1, N - 1)) \rceil$ bounds its distance envelope, and hence its distinct unique shortest paths, by $(2N)^k$.

First take generic input geometry. On each open angular phase with a unique unweighted assignment, relabel that assignment as the identity and apply [Proposition D.2](#). If the two defect labels coincide after relabeling, the diagonal plan is fixed instead. The vertex potential $|b_j|^2 - |b_i|^2$ telescopes from the residual costs. Split the angular circle at two antipodal chart seams; on each chart the remaining arc costs, divided by its positive cosine, are affine in the tangent parameter. By [Theorem 3.6](#), the unweighted problem has at most $N(N - 1)$ open phases; the two additional seams give at most $N(N - 1) + 2$ cells. Apply the path bound on each cell. A unique transport optimum at a boundary is strictly better than finitely many other vertices and therefore persists nearby, so it has already been counted.

For arbitrary input geometry, select one strict witness for each distinct unique transport plan. A sufficiently small generic perturbation of the two planar lists preserves all these finite strict gaps, without changing the marginals. The generic bound therefore applies to the original count. $\square$

These deductions leave open a support-polynomial bound, or a geometric superpolynomial lower construction, for arbitrary positive planar margins. Arbitrary affine-cost graph lower bounds cannot simply be imported: the geometric coefficients and all unintended complete-graph arcs must also be realized.

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